

Telomere-to-telomere genome assembly of a male pig provides insight into population structure and selection for body stature

Received: 23 September 2024

Accepted: 24 October 2025

Published online: 12 December 2025

 Check for updates

Ya-Biao Luo^{1,6}, Ning Huang^{1,2,6}, Cheng-Wan Zha^{1,6}, Li-Xian Yang¹, Peng-Xiang Xue¹, Qiao Xu^{1,3}, Xiao-Yang Yang¹, Long-Miao Zhang^{1,2}, Yu-Bei Wang^{1,2}, Zhe Chao⁴, Rui-Ping Sun⁴, Feng Wang⁴, Shan-Gang Jia⁵✉ & Mei-Ying Fang^{1,2}✉

A complete telomere-to-telomere (T2T) genome is essential for advancing pig genomic research. Here we assembled a pig T2T gap-free genome T2T-pig1.0 (2.63 Gb) for a boar in Wuzhishan, China, covering all 20 chromosomes. T2T-pig1.0 with an accuracy of >99.999% uncovers 194.42 Mb of previously unresolved regions (PURs), and 1,189 new genes are added to the current reference genome Sscrofa11.1. We annotated 111 protein-coding genes with 11 male-specific conserved orthologous genes on chromosome Y (43.25 Mb). Pig-specific centromeric satellite repeat units are revealed. Centromeric regions of all telocentric chromosomes harbor a unique structure, ‘telomere-SAT1B-(mSAT)_n-SAT3-q_arm’, and a few young long terminal repeats. With the addition of 339,092 single-nucleotide polymorphisms in PURs, the population structure is updated with cross-continental introgression, and a selective sweep analysis reveals 280 new regions and 133 new genes potentially associated with body stature. *GALNT13*, with strong selection signals, has a role in inhibiting porcine chondrocyte proliferation while promoting chondrocyte differentiation.

The domestic pig (*Sus scrofa*) has become an important farm and a biomedical animal worldwide, as it was domesticated approximately 10,000 years ago^{1,2}. Under long-term domestication and selection pressure, more than 200 domesticated pig breeds have emerged worldwide (<http://dad.fao.org/en/home.html>), and Wuzhishan pigs, a well-known pig breed native to China³, are popular in Hainan because of their unique meat flavor and small adult pig body size. Wuzhishan pig weight is approximately half that of popular commercial large-sized European pigs, such as Duroc, Yorkshire and Landrace pigs. The unique small body size of the Wuzhishan pig makes it an ideal model for studying

diabetes⁴, toxicity⁵, meniscus metamorphosis⁶ and other complex diseases, because of its lower feeding costs, lower rearing space requirements and easier handling. From domestication to modern breeding practices, humans have shaped the genomes of domestic pigs. Accurate and complete genome sequences are essential for promoting the applications of pigs in agricultural practice and biomedical research. To date, a total of 47 pig genome assemblies are available in the public NCBI database, with only 26 at the chromosome level, including the pig genome assembly Sscrofa9 and its updated versions Sscrofa10.2 (ref. 2) and Sscrofa11.1 (ref. 7; NCBI accession GCF_000003025.6).

¹State Key Laboratory of Animal Biotech Breeding, National Engineering Laboratory for Animal Breeding, MOA Key Laboratory of Animal Genetics and Breeding, College of Animal Science and Technology, Frontiers Science Center for Molecular Design Breeding, Beijing Key Laboratory for Animal Genetic Improvement, China Agricultural University, Beijing, China. ²Sanya Research Institute, China Agricultural University, Sanya, China. ³Jiang Xi Province Key Lab of Genetic Improvement, Institution of Biological Technology, Nanchang Normal University, Nanchang, China. ⁴Institute of Animal Science and Veterinary Medicine, Hainan Academy of Agricultural Science, Haikou, China. ⁵College of Grassland Science and Technology, China Agricultural University, Beijing, China. ⁶These authors contributed equally: Ya-Biao Luo, Ning Huang, Cheng-Wan Zha. ✉e-mail: shangang.jia@cau.edu.cn; meiying@cau.edu.cn

Table 1 | Comparison of pig genome assemblies Sscrofa11.1 and T2T-pig1.0

Feature	Sscrofa11.1	T2T-pig1.0
–	Genome summary	
Assembled bases (Gb)	2.53	2.63
Unplaced bases (Mb)	66.65	0
Gap bases (Mb)	29.86	0
Number of contigs	1,118	0
Contig N50 (Mb)	48.23	144.93
PUR (Mb)	–	194.42
–	Gene annotation	
Number of genes	20,803	21,713
–	SDs	
Percentage of SDs	3.23%	5.39%
SD bases (Mb)	81.82	141.93
Number of SDs	8,272	8,238
–	RepeatMasker	
Percentage of repeats	44.28	48.25
Repeat bases (Mb)	819.37	1,270.55
LINEs	609.32	643.95
SINEs	84.61	89.16
LTRs	125.44	137.67
Satellite	32.35	50.52
DNA	85.34	145.93
Simple repeat	20.51	31.90

However, these draft genomes have tens of thousands of gaps on the chromosomes, for example, 5,323 gaps in Sscrofa10.2 and 506 gaps in Sscrofa11.1.

Recent breakthroughs in the high-resolution sequencing of highly repetitive and complex regions have enabled complete telomere-to-telomere (T2T) genome assembly, which has led to the resolution of all the gaps, such as those in human⁸, sheep⁹, goat¹⁰ and mouse¹¹ genomes, and similar progress has also been achieved in pigs¹²; these T2T assemblies have greatly expanded the potential for comprehensive genomic research. In this study, we present a T2T gap-free genome assembly (T2T-pig1.0) comprising 18 autosomes and chromosomes X (chrX) and Y (chrY) for a Wuzhishan boar. The high continuity and completeness of T2T-pig1.0 enable precise genomic analyses of centromeric structures, previously unresolved regions (PURs), cross-species comparisons on chrX/chrY and structural variations (SVs). Because of the newly added single-nucleotide polymorphisms (SNPs), the genetic structure of the pigs worldwide was reconstructed and the new selection signals and genes related to small stature were successfully identified. This genomic resource and our findings are valuable for further improvements of this important livestock species, and for expanding the potential of pigs as biomedical models^{2,13}.

Results

T2T assembly of all 20 chromosomes

We selected a Wuzhishan boar for T2T genome assembly and generated 169.8 Gb of ultralong Oxford Nanopore Technology (ONT) data (read N50 of 108.9 kb, 65.31× coverage), 192.4 Gb of PacBio HiFi data (read N50 of 20.4 kb, 73.98× coverage), 1,113.8 Gb of Bionano optical mapping data (≥150 kb), 267 Gb of chromosome conformation capture sequencing (Hi-C) data (100× coverage) and 133.7 Gb (50× coverage) of next-generation sequencing (NGS) data (Supplementary Table 1).

The contigs were assembled from the ultralong ONT and HiFi reads using NextDenovo¹⁴ and Hifiasm¹⁵, and given the accuracy of the HiFi reads, the Hifiasm contigs were selected for subsequent assembly and validated with Bionano data (Supplementary Fig. 1). We anchored all the contigs onto 20 pseudochromosomes (18 autosomes + X + Y) using Hi-C data, which corresponded to the chromosomes and their orientations in Sscrofa11.1, and this preliminary chromosome-level assembly with a total of 2.6 Gb contained 104 contigs and 21 gaps. The gaps were then filled based on the ONT data and polished using long and short reads (Supplementary Fig. 2). Using a sequence query for telomeric repeats (TTAGGG or CCCTAA), we assembled all 40 telomeres spanning 0.42 Mb and placed them on both ends of 20 T2T chromosomes (Supplementary Table 2). Finally, the gap-free T2T genome assembly T2T-pig1.0 was generated with a size of 2.63 Gb (Table 1).

We assessed and confirmed the completeness and accuracy of T2T-pig1.0 in multiple ways (Supplementary Text), including the read alignment rate (Extended Data Fig. 1), Hi-C contact maps (Extended Data Fig. 2), Bionano optical maps (Extended Data Fig. 3), the embryophyte benchmarking universal single-copy orthologs (BUSCOs¹⁶; Supplementary Fig. 4) and quality values (QVs) across all chromosomes (Supplementary Table 3). T2T-pig1.0 contained 194.42 Mb of PURs, which were not assembled on the chromosomes of Sscrofa11.1 (Fig. 1a). Although T2T-pig1.0 exhibited high collinearity with Sscrofa11.1, 12,358,943 single-nucleotide variations, 183 duplications, 215 inversions and 97 translocations were identified between these two assemblies (Extended Data Fig. 4). For example, we observed one of the longest inversions (chr10–SV1, 25.3 Mb) located on the short arm of chr10, which showed three types, that is, T2T-pig1.0-like, Sscrofa11.1-like and others (Extended Data Fig. 5).

In addition, we used Hi-C, ONT and HiFi reads to assemble the haplotypic chromosomes and identified the heterozygous regions in assembly graph of T2T-pig1.0 (Supplementary Fig. 3). The haplotype-resolved contigs for Hap1 (2.59 Gb) and Hap2 (2.45 Gb) were obtained with 51 and 38 gaps, respectively, that did not completely correspond to the maternal and paternal contigs because parental sequencing data were not available. A total of 1,805 SVs (>1 kb) were identified in the heterozygous regions of the haplotypes in T2T-pig1.0 (Supplementary Fig. 3).

Genome annotation

A combination of de novo and homology-based methods was used to annotate a total of 1.27 Gb of repetitive elements, representing 48.25% of the entire T2T-pig1.0 genome (Fig. 1c and Table 1). These repeats consisted of 24.46% long interspersed nuclear elements (LINEs), 5.54% DNA transposons, 5.23% long terminal repeats (LTRs), 3.39% short interspersed nuclear elements (SINEs), 1.92% satellites and 7.71% other repeats (Supplementary Text, Supplementary Fig. 5 and Supplementary Table 4). We characterized 8,238 nonredundant (NR) segmental duplications (SDs) in the T2T-pig1.0 assembly spanning 141.93 Mb, with enriched SDs on chr3 and chr10 (Supplementary Fig. 6 and Supplementary Table 5), and 273 SDs were newly identified in T2T-pig1.0. In contrast, 8,272 SDs (81.82 Mb) were identified in the current reference genome, Sscrofa11.1.

In T2T-pig1.0, we annotated a total of 38,306 genes, including 21,713 protein-coding genes, 2,385 pseudogenes, 7,301 long noncoding RNAs (lncRNAs), 828 microRNAs (miRNAs), 937 small nuclear RNAs (snRNAs), 4,447 transfer RNAs (tRNAs) and 695 ribosomal RNAs (rRNAs; Table 1, Supplementary Table 6 and Supplementary Figs. 7 and 8). The protein-coding genes were of high quality, with 99.3% (21,561) genes functionally annotated, 92.83% (20,156) transcriptionally supported, and a BUSCO completeness score of 97.9% (Fig. 1d). Among these genes, 1,189 protein-coding genes in T2T-pig1.0 were located in the PURs compared with the Sscrofa11.1 reference genome, and these PUR genes exhibited diverse expression profiles across 15 tissues, especially in the hypothalamus and pituitary gland (Supplementary Fig. 9a).

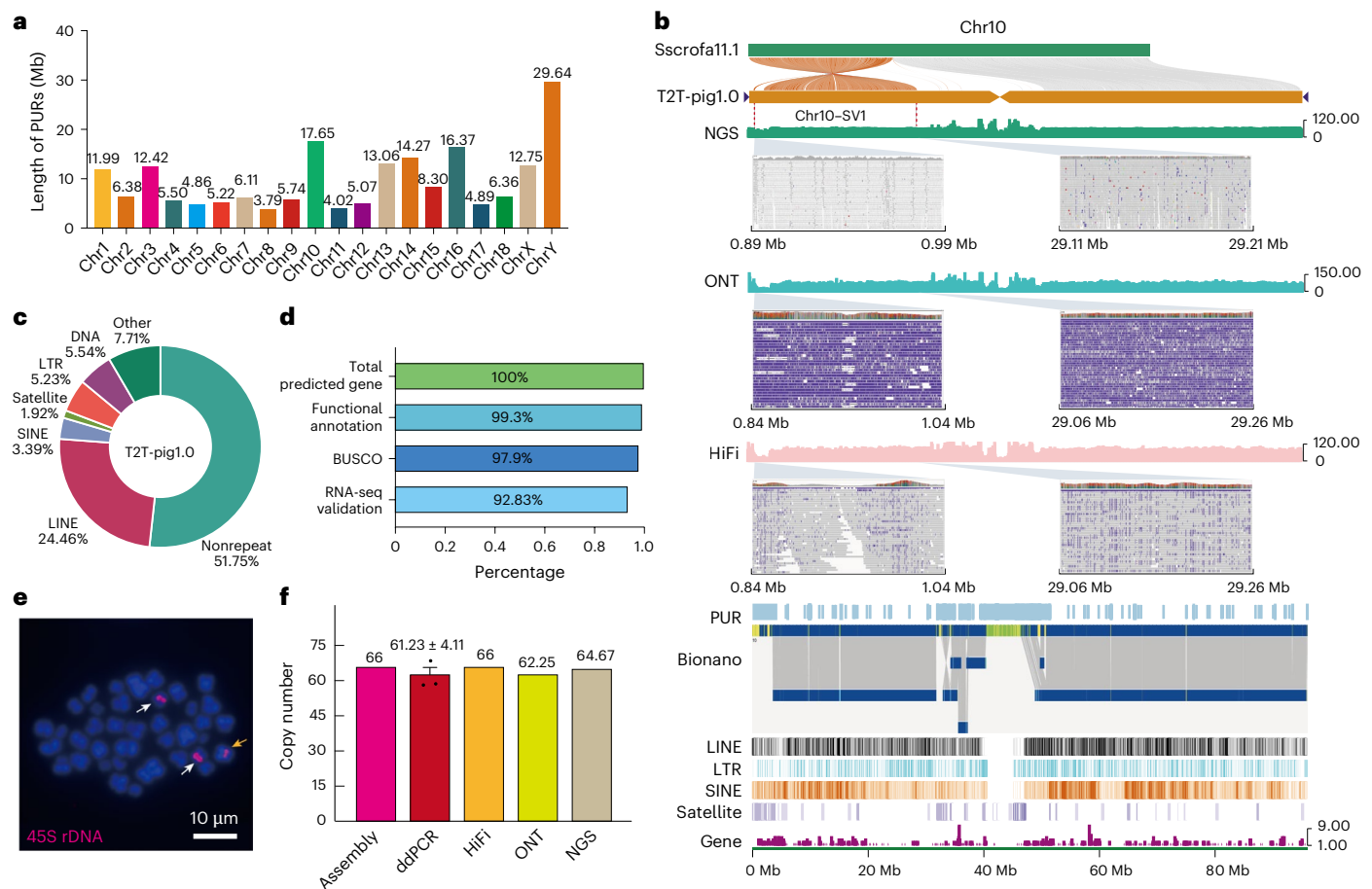


Fig. 1 | Summary of the complete T2T-pig1.0 pig genome assembly. a, Length of PURs per chromosome in T2T-pig1.0 relative to Sscrofa11.1. **b**, Schematic representation of features on chr10 in T2T-pig1.0. From top to bottom, the collinearity (gray blocks) between T2T-pig1.0 (orange bar, with black triangles for telomeres) and Sscrofa11.1 (sea green bar) with inversions in red blocks (including the largest one chr10–SV1); coverage and IGV snapshots on the joint sites of chr10–SV1 of NGS, ONT and HiFi reads from the T2T-pig1.0 individual; PURs in light blue; Bionano optical mapping, blue bars for the collinearity linked with gray blocks, green and yellow bars for the break of collinearity; and density values for LINE, LTR, SINE, satellite and gene. **c**, Repeat composition of the whole T2T-pig1.0

genome. **d**, Genome annotation for the protein-coding genes in T2T-pig1.0. From top to bottom—the total number of predicted protein-coding genes, the number of genes with functional annotations against the public databases, the BUSCOs and the number of genes supported by the RNA-seq data. **e**, FISH imaging using probes of 45S rDNA. The orange and white arrows indicate chr8 and chr10, respectively. The experiments were repeated independently at least thrice with similar results. **f**, Copy number of 45S rDNAs in the assembly (T2T-pig1.0) was estimated on the basis of the coverages of PacBio HiFi, ONT, NGS reads and ddPCR with three technical replicates, and the experiments were repeated thrice. The copy number values are shown on the top of the bars, with mean ± s.d. for ddPCR.

We further found that the gene family members in PURs experienced lower evolutionary pressure due to the substantially lower nonsynonymous (K_a) and synonymous (K_s) substitution rates (K_a/K_s , $P < 0.001$; Supplementary Fig. 9b) than their paralogous genes in non-PURs.

The rDNA arrays present a challenge for complete genome assembly, and T2T-pig1.0 resolved the 45S rDNA regions with 66 copies on chr8 and chr10, which is consistent with the results of previous in situ hybridization studies¹⁷. Our fluorescence in situ hybridization (FISH) results further confirmed the binding of 45S rDNA probes to the three chromosomes (Fig. 1e), rather than the four chromosomes for chr8 and chr10, which could be explained by the haplotypic deletion of 45S rDNA on chr8 (ref. 17). The 45S rDNA copy number estimated based on the sequences of chr8 and chr10 in T2T-pig1.0 was further verified by digital droplet PCR (ddPCR) and the average coverage of ONT, HiFi and NGS reads (Fig. 1f).

Complete genomic structure of porcine centromeres

We determined centromeric regions on all chromosomes, with a total length of 107.3 Mb (4.0% of the T2T-pig1.0 genome), through a combination of chromatin immunoprecipitation followed by sequencing (ChIP-seq), HiFi methylation and centromeric satellite signals

(Fig. 2a,b and Supplementary Fig. 10). Based on the locations of the centromeric regions, 12 metacentric/submetacentric chromosomes (chr1–chr5, chr8–chr12, chrX and chrY), 2 subtelocentric chromosomes (chr6 and chr7) and 6 telocentric chromosomes (chr13–chr18) were determined and largely consistent with previously published karyotype results¹⁸. We observed pronounced DNA hypermethylation across the centromeric regions of all chromosomes, with the exception of a small hypomethylated zone known as the centromeric dip region (CDR). In mammals, the centromeric histone H3 variant (CENP-A) is an essential component for kinetochore assembly¹⁹. ChIP-seq analysis further confirmed the centromeric locations, which were correlated with the CDRs²⁰.

Centromeric regions are largely composed of satellite repeat units. A total of seven centromeric satellite units were identified on the autosomes and chrX and ultimately categorized into three groups based on the sequence similarity (Supplementary Text, Supplementary Fig. 11 and Supplementary Table 7), that is, SAT1 (SAT1A = 336 bp, SAT1B = 336 bp and SAT1C = 334 bp), SAT2 (499 bp) and SAT3 (SAT3A = 219 bp, SAT3B = 408 bp and SAT3C = 687 bp). We found that the distribution of centromeric satellite units exhibited a strong distributional bias (Fig. 2c,d). Our FISH results confirmed the

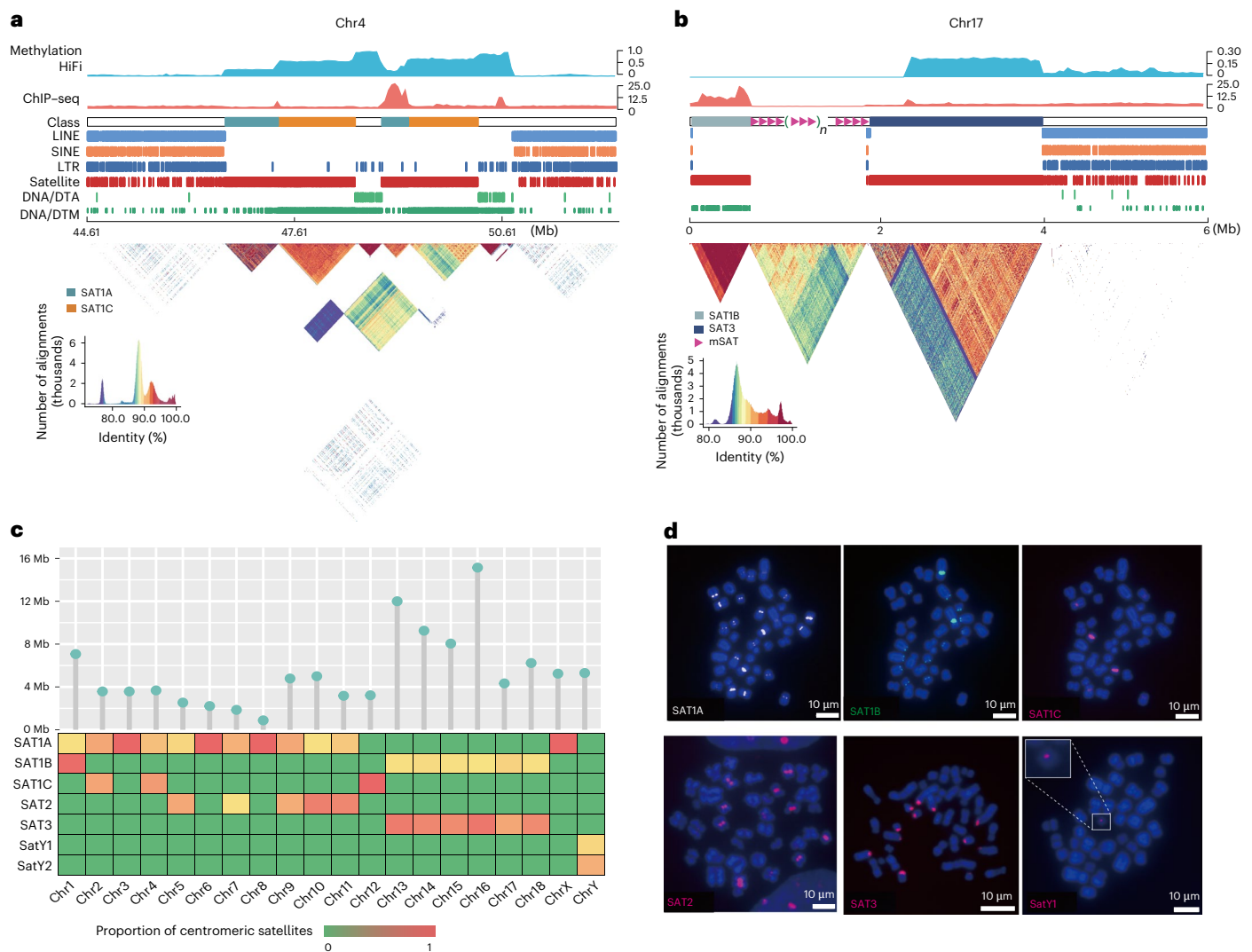


Fig. 2 | Centromeric structure in T2T-pig1.0 pig genome assembly. a,b, The centromeric region structures on metacentric chr4 (**a**) and telocentric chr17 (**b**) are shown. From top to bottom—methylation by HiFi reads, ChIP-seq for CENP-A protein enrichment through CENP-A antibodies in 5-kb windows, classification of the centromeric repeats (including SAT1A, SAT1B, SAT1C, SAT2 and microsatellites) for mSAT, LINE, SINE, LTR, satellites, DNA/DTA and DNA/DTM. The pairwise 10-kb sequence identity (%) heatmap in the centromeric

region is shown below. **c,** Length distribution of centromeric regions (top) and proportions of centromeric satellite units (SAT1A–SAT1C, SAT2, SAT3, SatY1 and SatY2) on all the autosomes and chrX and chrY (bottom). **d,** FISH imaging using probes for SAT1A–SAT1C, SAT2, SAT3 and SatY1. SatY1 is shown uniquely on the chrY. Bottom-right, the enlarged image of SatY1 probe (pink) and its unique binding to the chrY indicated by white lines. The experiments were repeated independently at least thrice with similar results.

presence and distribution of these centromeric satellite units across the chromosomes (Fig. 2d and Extended Data Fig. 6). SAT3 was exclusively localized in the centromeric regions of six telomeric chromosomes (Fig. 2c and Supplementary Fig. 10), together with SAT1B. Intriguingly, a unique microsatellite sequence of ‘AGAAT’ (mSAT) was substantially enriched in the CDRs of all six telocentric chromosomes, and exhibited a specific distribution pattern of ‘telomere-SAT1B-(mSAT)_n-SAT3-q_arm’ (Fig. 2b and Supplementary Fig. 10). To the best of our knowledge, microsatellites have not been reported in the centromeric regions of T2T assemblies from other animals, such as humans²¹, goats¹⁰ and sheep⁹, although they are found in the centromeric region of chr2 in *Drosophila*²². Following the overall increases of DNA transposons in T2T-pig1.0 (Table 1), we investigated the mutator-like (DTM) and hAT (DTA) families in the centromeric regions, which account for approximately 60% of the total DTM/DTA length in the assembly (Fig. 2a,b and Supplementary Fig. 10). In the centromeric regions of telomeric chromosomes, DTAs were absent, whereas DTMs were found exclusively in the SAT1B-enriched domains.

We screened the whole genome of T2T-pig1.0 for LTRs and estimated their insertion time based on the intact LTRs and their mutations. According to the distribution of insertion time, we found that the centromeric regions underwent substantial invasion by LTRs that were predominantly young (<15 million years ago (MYA); Extended Data Fig. 7a). The average insertion time of these centromeric LTRs was 12.31 MYA, which was noticeably younger than the genome-wide average of 41.11 MYA (Extended Data Fig. 7b). The rapid expansion of young LTRs in centromeric regions of T2T-pig1.0 suggests that LTRs may be involved in the evolution of centromeres in pigs, and further indicates that centromeres with frequent LTR invasion may represent recent evolutionary footprints. However, in contrast to frequent young LTR invasions observed in the centromeric regions of telomeric chromosomes in humans, goats and sheep, centromeric regions of porcine telocentric chromosomes exhibited a strikingly low level of invasion by young LTRs. (Extended Data Fig. 7a). In summary, the centromeric repeats in T2T-pig1.0, including DNA transposons, microsatellites and LTR invasion, are not identical as described in the available T2T genome assemblies of humans, goats and sheep.

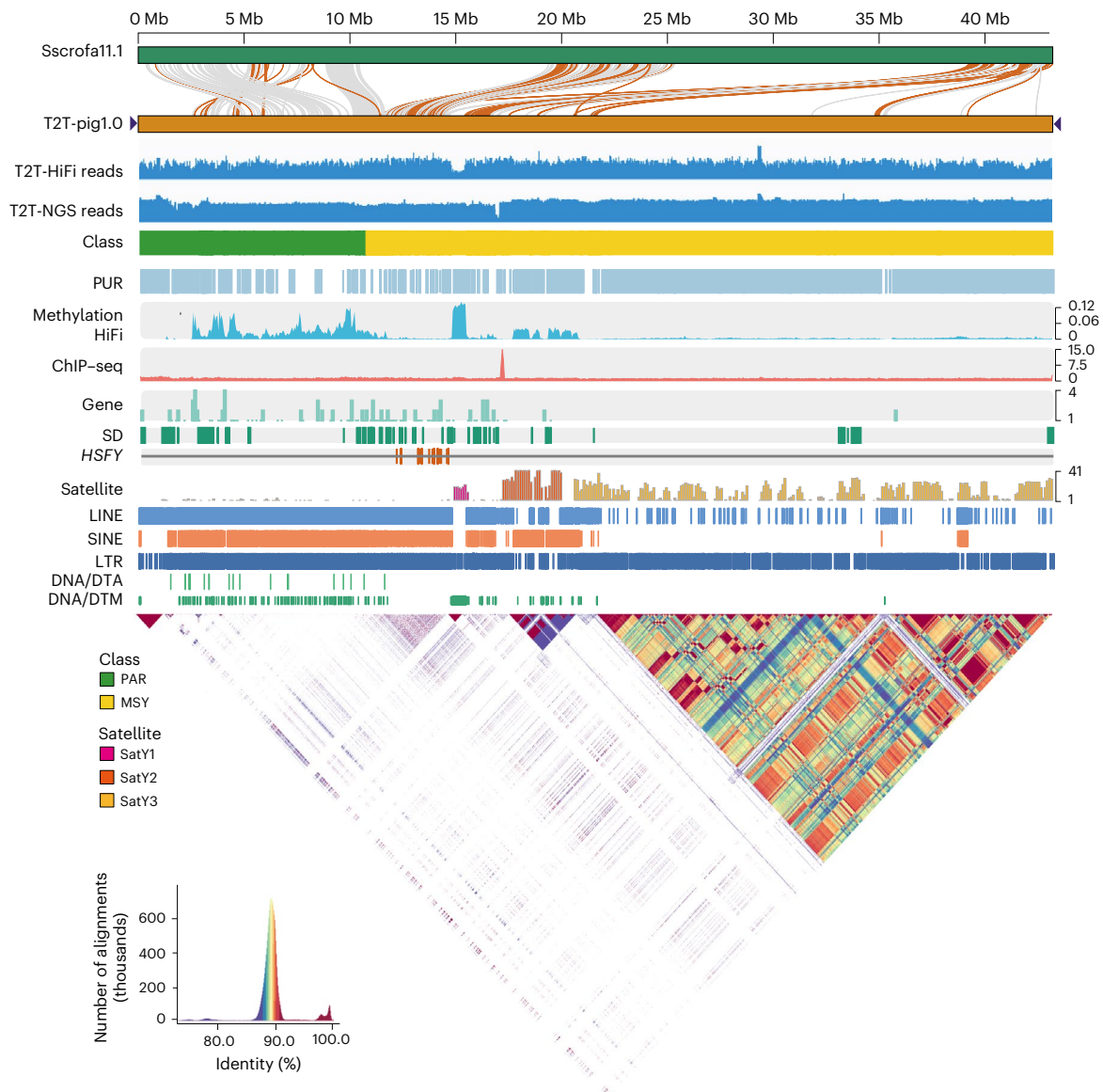


Fig. 3 | ChrY structure and features. Features of the chrY in T2T-pig1.0. From top to bottom—collinearity of chrY between Sscrofa11.1 and T2T-pig1.0 with telomeres indicated by two blue triangles, class for the two regions of PAR and MSY, PURs in T2T-pig1.0 relative to Sscrofa11.1, methylation (5 mC) levels estimated with HiFi reads, ChIP-seq signals, gene density, SDs, *HSFY* genes, satellite density (including SatY1, SatY2 and SatY3 (yellow)), LINE density, SINE density, LTR density, DNA/DTA and DNA/DTM. The pairwise 10-kb sequence identity (%) heatmap across the chrY of T2T-pig1.0 is shown below. mC, methylcytosine.

satellite density (including SatY1, SatY2 and SatY3 (yellow)), LINE density, SINE density, LTR density, DNA/DTA and DNA/DTM. The pairwise 10-kb sequence identity (%) heatmap across the chrY of T2T-pig1.0 is shown below. mC, methylcytosine.

ChrY structure

The T2T gap-free Y assembly in T2T-pig1.0 with a size of 43.25 Mb reveals 29.64 Mb of previously uncharacterized sequences compared to the chrY sequences (43.55 Mb with ~28 Mb unknown 'N' bases) of Sscrofa11.1 (ref. 23; Fig. 3 and Supplementary Table 8). Based on the alignment of chrX and chrY, we determined the pseudoautosomal region (PAR) and the male-specific region of the chrY (MSY; Extended Data Fig. 8). We annotated a total of 111 protein-coding genes on chrY of T2T-pig1.0, 42 of which were newly identified in the PURs (Fig. 3). The annotation revealed two or more copies of ten genes on chrY, including 14 copies of the heat shock transcription factor Y-linked like (*HSFY*; Supplementary Table 8). The differences in *SHOX* (2 versus 1) and *HSFY* (25 versus 14) gene copy number on chrY between Sscrofa11.1 and T2T-pig1.0 could be explained by breed variation. Compared with the 10.38-Mb repeat sequences detected on chrY in Sscrofa11.1, we further identified a total of 36.72-Mb (84.9%) repeat sequences on chrY in T2T-pig1.0 (Supplementary Table 8), whose proportion is identical

to 84.9% on human chrY²⁴. In addition, similar to the human chrY, the q arm of chrY in T2T-pig1.0 was dominated by a large region of satellite arrays (20.98 Mb), which consisted of the repeat unit SatY3. SatY3, which is 3,943 bp in length (Supplementary Table 7), is highly similar to a male-specific DNA repeated sequence (3,832 bp, accession [X12696.1](#)) previously published in domestic pigs²⁵. LINES, LTRs and SINEs were enriched on the p arm of chrY (Fig. 3).

The centromeric repeat units SAT1, SAT2 and SAT3 were not found on chrY of T2T-pig1.0. A ChIP-seq analysis based on antibodies targeting CENP-A revealed a prominent enrichment peak spanning the 17.13–17.21 Mb region on chrY (Fig. 3), which coincides with a hypermethylation region spanning approximately 14.75 Mb to 20.82 Mb. Furthermore, in this hypermethylation region, we identified two putative centromeric repeats of SatY1 (5,660 bp) and SatY2 (2,472 bp; Supplementary Table 7), which showed high sequence identity in the pairwise sequence identity heatmap (Fig. 3). FISH imaging confirmed the specific localization of SatY1 to chrY (Fig. 2d).

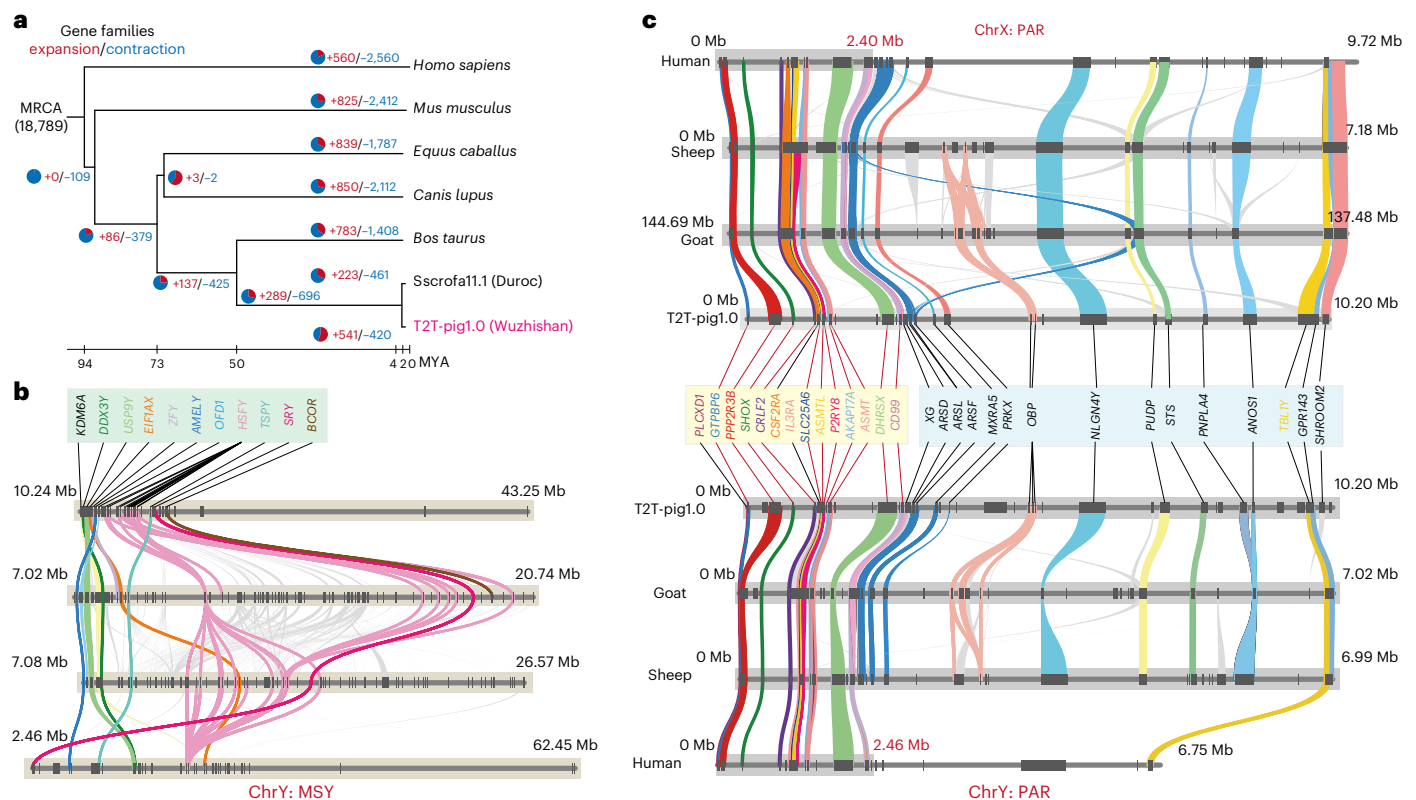


Fig. 4 | Comparative analysis of chrY in T2T-pig1.0 and other three T2T genome assemblies. **a**, Phylogenomic analysis and expanded and contracted gene families in the seven genomes, including pigs (T2T-pig1.0 and Sscrofa11.1), and other five animal species (humans, *Homo sapiens*, GCF_009914755.1; mice, *Mus musculus*, GCF_000001635.27; horses, *Equus caballus*, GCF_002863925.1; wolves, *Canis lupus*, GCF_011100685.1; and cattle, *Bos taurus*, GCF_002263795.3). The values on the line below represent the divergence time from the present (MYA). The MRCA is placed as the root. **b**, The MSY is compared among the T2T chrY of pigs (T2T-pig1.0), goats (T2T-goat1.0), sheep (T2T-sheep1.0) and humans

(T2T-CHM13v2.0). The 11 orthologous genes are shown with colored gene symbols in a green box, and shared by all these four species. The genes are shown as small black bars on the horizontal chrY. **c**, The PARs on chrX (top) and chrY (bottom) are compared for orthologous genes across the four species. The gray box covers the PARs on the gray lines for chrX and chrY. The 12 genes in the PAR on chrY of humans are lost, and shown with black gene symbols in a light blue box. The genes shared by all four species are shown in colored gene symbols in the PARs of chrY. MRCA, most recent common ancestor.

Cross-species comparative genomic analysis

A total of 18,789 gene families were identified across the genome assemblies of six mammalian species, that is, humans, mice, horses, wolves, cattle and pigs (including both Sscrofa11.1 and T2T-pig1.0). Based on the single-copy orthologs identified among the seven genomes, we constructed a phylogenetic tree, which revealed that Wuzhishan and Duroc pigs diverged approximately 2.1 MYA (confidence interval = 0.8–5.5 MYA). We identified 541 gene families that were expanded in T2T-pig1.0, compared with 223 in Sscrofa11.1 (Fig. 4a). The Gene Ontology (GO) analysis revealed substantial enrichment of these expanded genes in diverse biological pathways. For instance, terms such as ‘negative regulation of cell size’ and ‘negative regulation of organ growth’ (false discovery rate < 0.05; Supplementary Table 9) may be associated with smaller body stature.

A high degree of gene synteny was observed across the four species (humans²⁴, goats¹⁰, sheep⁹ and pigs) with T2T genomes (Extended Data Fig. 9), although their chromosomes underwent extensive breakages and fusions. We focused on the comparison of sex chrX and chrY across the four animal T2T genome assemblies. A distinct heterochromatic region composed of tandem repeat arrays is always found at the end of the chrY q arm, including SatY3 in pigs, HSat3 in humans and ZFY gene in sheep, which might be related to the inhibition of X–Y recombination during meiotic pairing. Nevertheless, goats appear to lack such a repeat array at the end of chrY. In the MSY region, we identified 11 chrY orthologous genes (YOGs) across the four species, including known male-specific genes *SRY*, *TSPY* and *HSFY*, indicating

high evolutionary conservation (Fig. 4b and Supplementary Table 10). PARs were also compared between chrX and chrY in the four animals (Fig. 4c). Based on the alignment of chrX and chrY, the longest PAR on chrY in pigs was ~10.20 Mb, while the PARs on chrY in sheep and goats were approximately 6.99 Mb and 7.02 Mb, respectively. Humans have lost 15 orthologous genes from *XG* to *SHROOM2*, while maintaining *TBL1Y* on chrY, which are shared by both the chrX and chrY of the other three species. Consequently, humans possess small PARs of approximately 2.46 Mb and 0.33 Mb, respectively. Furthermore, 12 and 13 orthologous genes were identified in the PARs of chrX and chrY, respectively, across the four species (Fig. 4c). Among these genes, 11 orthologous genes in the PARs, namely, *GTPBP6*, *PPP2R3B*, *SHOX*, *CRLF2*, *SLC25A6*, *ASMTL*, *P2RY8*, *AKAP17A*, *ASMT*, *DHRX* and *CD99*, could be found on both chrX and chrY, and have not been shown to be related to the male-specific functions. For instance, *ASMTL*, *P2RY8* and *CD99* have been implicated primarily in the immune response^{26–28}, and *SHOX* is involved in skeletal development²⁹. The high synteny based on orthologous genes in PARs and MSYs indicates a single evolutionary origin of chrX and chrY across these large animals.

SV calling based on long reads

We used T2T-pig1.0 as a reference to call 83,106 large SVs based on ONT long reads with an average read length of 21.9 kb and a read N50 value of 44.9 kb from ten samples (Supplementary Table 11). T2T-pig1.0 showed advances in SV calling in contrast to the Sscrofa11.1 used as a reference (Supplementary Text and Supplementary Fig. 12). For instance, the

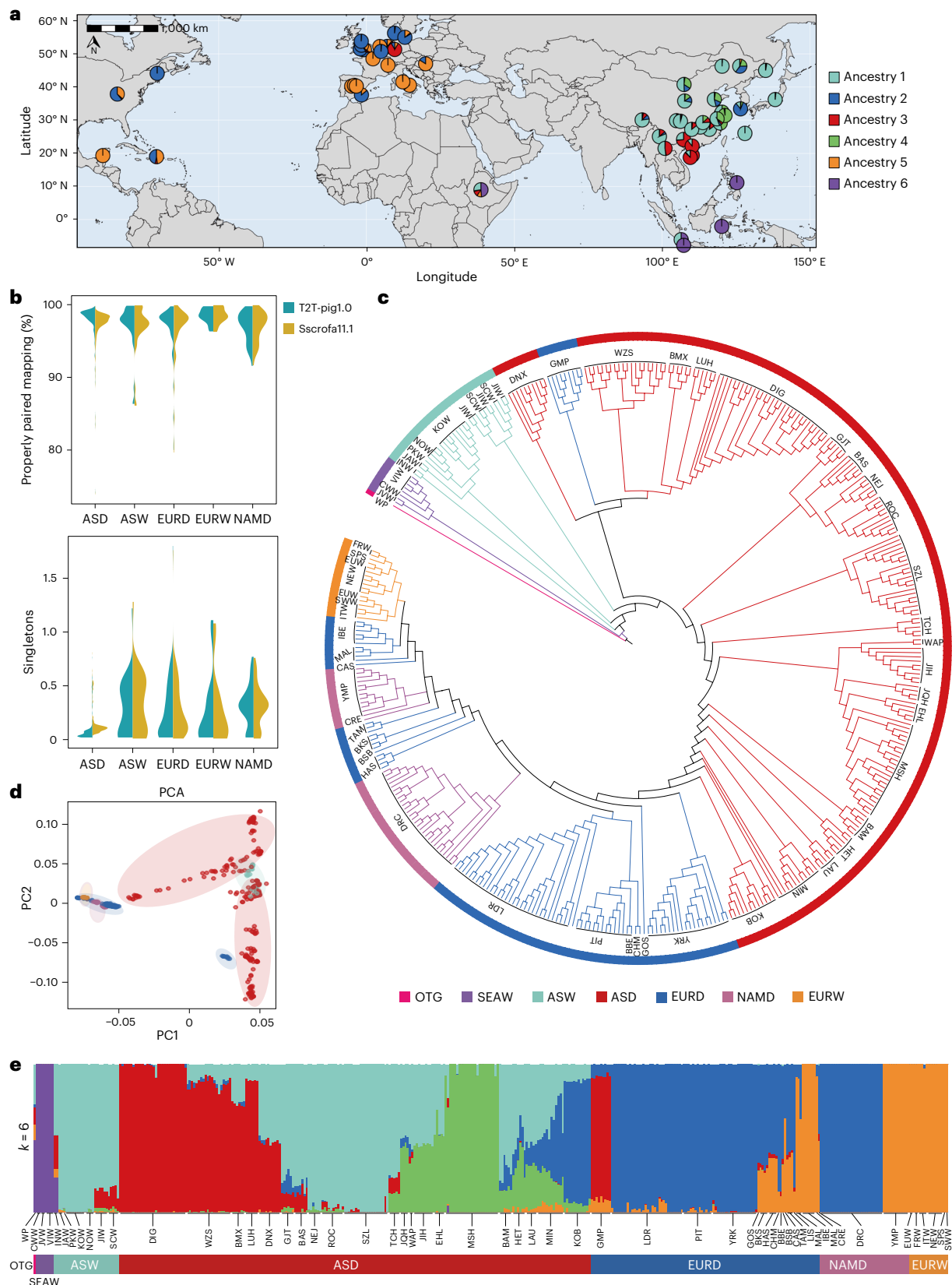


Fig. 5 | Improvements in variant calling and global population structure. **a**, Geographic distributions of domestic and wild pigs in this study. The pig samples are colored on the basis of their six ADMIXTURE ancestries (1–6). The map was generated using the open-source R packages *mapmixture* (v.1.2.0)⁵⁹. **b**, The alignment of NGS reads is improved for mapped paired reads and singletons in T2T-pig1.0 compared with Sscrofa11.1. **c**, **d**, The NJ tree (c) of wild

and domestic pigs and PCA (d) of five geographic pig populations were generated on the basis of LD-pruned SNPs. **e**, Population genetic structure of wild and domestic pigs inferred from ADMIXTURE ($k = 6$) using T2T-pig1.0 as the reference. The breed names are included in Supplementary Table 12. PCA, principal component analysis; N, neighbor-joining.

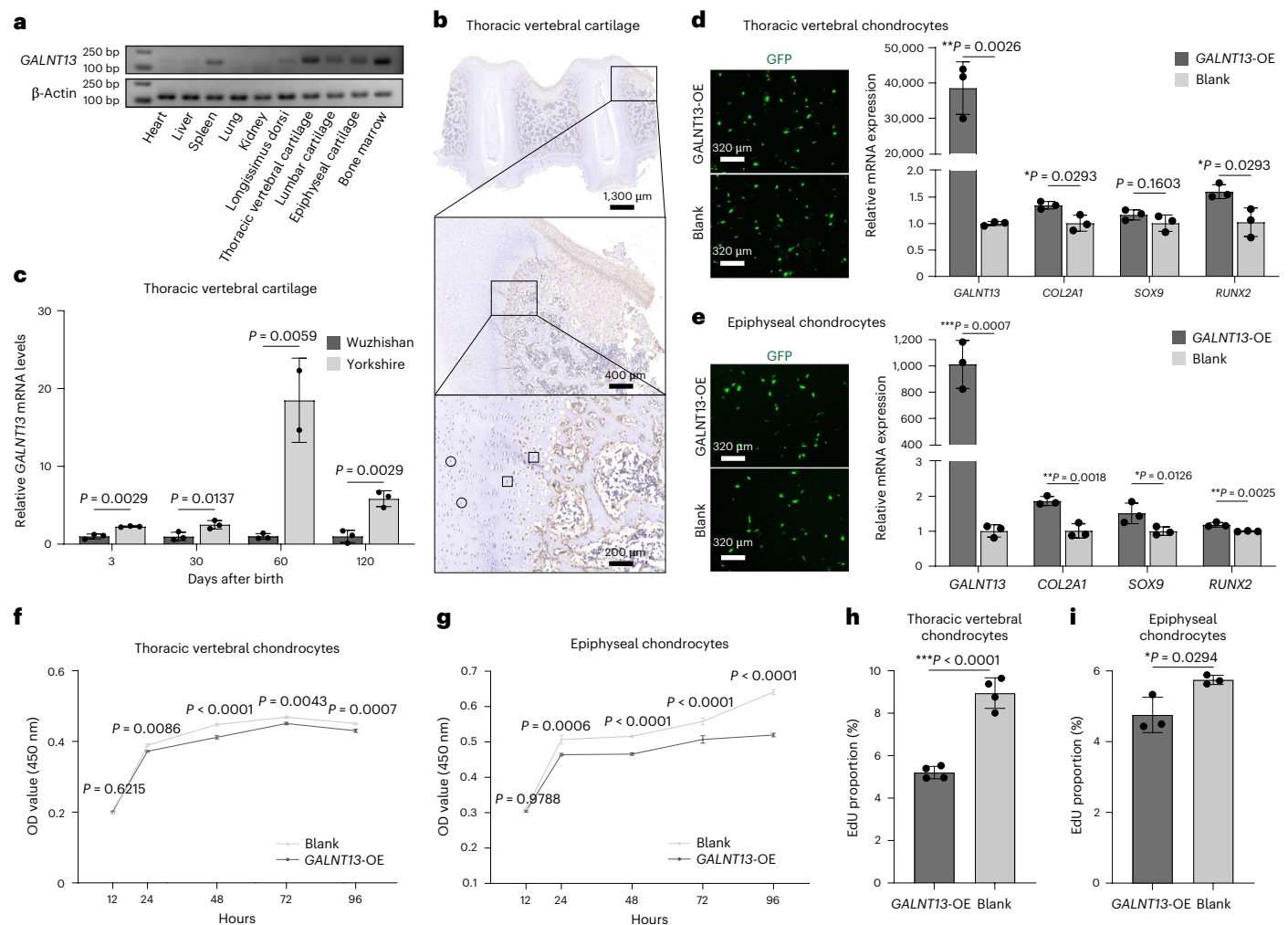


Fig. 6 | Association of *GALNT13* with pig chondrocyte development.

a, *GALNT13* mRNA expression in ten tissues of Wuzhishan pigs, with β -actin as the control gene. **b**, Immunohistochemical observation of the *GALNT13* protein expression in porcine thoracic vertebral body tissue at 3 days of age. The boxed and circled areas highlight *GALNT13*-positive cells, corresponding to round and hypertrophic chondrocytes, respectively. The experiments in **a** and **b** were repeated independently at least three times with similar results. **c**, *GALNT13* mRNA expression levels in thoracic vertebral cartilage. Expression levels of *GALNT13* in Wuzhishan pigs at the same time point were normalized to 1. Data are presented as mean $\pm$ s.d. from three biological replicates, except for one data point (60-day Y) with two biological replicates. **d, e**, The expression levels of the chondrogenic marker genes *COL2A1*, *SOX9* and *RUNX2* in primary thoracic chondrocytes (**d**) and epiphyseal chondrocytes (**e**) are shown for

5 days of chondrogenic differentiation after transfection with the *GALNT13* overexpression (*GALNT13*-OE) vector and the blank control vector (blank). The GFP imaging is shown in the left. **f–i**, Conventional culture of primary thoracic chondrocytes and epiphyseal chondrocytes via the CCK-8 assay (**f**, $n = 5$, and **g**, $n = 5$) and the EdU cell proliferation assay (**h**, $n = 4$, and **i**, $n = 3$) showed the inhibition of *GALNT13* overexpression (*GALNT13*-OE). Data in **d–i** are presented as mean $\pm$ s.d. Differences highlighted in **c–e**, **h** and **i** were assessed by unpaired, two-tailed Student's *t* tests. Optical density (OD) values at 450 nm were obtained for cell proliferation in the CCK-8 assay. Multiple comparison correction was performed by controlling the false discovery rate at 1.0% with the two-stage step-up method of Benjamini, Krieger and Yekutieli. Statistical analysis in **f** and **g** was performed by two-way analysis of variance with Sidak's multiple comparison test ($\alpha = 0.05$). * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

number of SVs increased by 46.92% and 55.84% in European domestic pigs and wild boars, respectively. A total of 27,055 SVs identified by T2T-pig1.0, including 15,998 deletions and 10,912 insertions, are located in PURs.

Population structure and divergence

We collected whole-genome sequencing datasets from a total of 407 samples (Supplementary Table 12) from 60 breeds, including domestic pigs, commercial pigs and wild boars, on three major continents, namely, Asia, Europe and North America (Fig. 5a). The number of paired reads correctly mapped to T2T-pig1.0 was substantially increased compared with that to Sscrofa11.1, whereas the mismatch rate decreased with fewer singletons (Fig. 5b and Supplementary Fig. 13). We detected a total of 17,846,955 and 15,835,033 SNPs with T2T-pig1.0 and Sscrofa11.1, respectively, as references. Among them, a total of 339,092 SNPs were identified in the PURs, accounting for 1.9% of the total SNPs.

A neighbor-joining tree (Fig. 5c) was constructed by using all the SNPs identified in T2T-pig1.0, with *Phacochoerus africanus* as an out-group (OTG). In the tree, principal component analysis (Fig. 5d) and ADMIXTURE analysis (Fig. 5e), all samples were classified into six major groups, including Asian domestic pig (ASD), European domestic pig (EURD), North American domestic pig (NAMD), Asian wild pig (ASW), European wild pig (EURW) and Southeast Asian warty pig (SEAW). Rare alleles (allele frequency of <0.05) were identified among the SNPs located in the PURs, and were dominant in the ASD, ASW, EURD and SEAW (Extended Data Fig. 10a,b). Moreover, most SNPs in PURs were fixed or near-fixed alleles (allele frequency of >0.95) in EURW and NAMD. Previous studies have classified ASD into six geographically distinct subclades corresponding to South China, the Tibetan Plateau, Southwest China, East China, Central China and North China³⁰. In contrast, we revealed three major ancestries associated with ASD (Fig. 5a,e),

that is, ancestry 1 dominating in ASW, ancestry 3 in South China and ancestry 4 in East China, consistent with previous results³¹. European ancestries (ancestry 2 for EURD) could also be found among pigs from Northeast Asia (Fig. 5a). These observations with T2T-pig1.0 as a reference revealed genetic introgression among the Asian pig populations, and improved the dissection of pig genetic structure compared with those using Sscrofa11.1 as a reference (Supplementary Fig. 14).

Selection signatures related to body stature

Body size is one of the most economically substantial traits in swine production, with certain breeds exhibiting more than a twofold difference in mature weight. For example, compared with wild boars and commercial large pigs, Wuzhishan and Bamaxiang pigs are well-known miniature breeds that exhibit substantially smaller body sizes. Therefore, we selected Wuzhishan ($n = 20$) and Bamaxiang ($n = 6$) pigs, along with large commercial pigs (Duroc, $n = 29$; large white, $n = 20$; Supplementary Table 12), to identify selective sweep signals based on the SNPs and the top 5% of the F_{ST} values (Supplementary Fig. 15a). Through pairwise comparisons among Yorkshire versus Wuzhishan, Yorkshire versus Bamaxiang, Duroc versus Wuzhishan and Duroc versus Bamaxiang pigs, we identified 51,204 to 332,121 selected SNPs (Supplementary Table 14). By overlapping these SNPs, we obtained 2,956 NR-selected regions corresponding to 657 candidate genes (permutation test, $P = 0.00009$), with 280 in PURs. Interestingly, 331 of the 657 selected genes (50.4%) are clustered within the 40–135-Mb region of chrX, and 248 of the top 250 selected genes are located within this sweep region (Supplementary Table 13). The striking selection signatures on chrX attributed to the extremely low rate of recombination in this region³². Using T2T-pig1.0 as a reference, 133 selected genes were uniquely identified by overlapping all four comparisons, in contrast to Sscrofa11.1 (Supplementary Fig. 15b), and exhibited diverse expression patterns across 15 tissues (Supplementary Fig. 15c), with particularly high expression levels in the pituitary gland, hypothalamus, spleen and pancreas.

We found that some selected genes affect pig body size, including *PLAG1* (ref. 33) and *NR6A1* (ref. 34), which have also been reported to be associated with body stature in other species, such as snakes³⁵, cattle³⁶ and sheep³⁷. Strikingly, three genes, *BRCA1*, *CORIN* and *GALNT13*, are collectively under strong selection according to both the F_{ST} values and π ratios across all four large-versus-small comparisons (Supplementary Fig. 16), and are known to influence body stature and bone development in humans and mice^{38–40}. The selected SNPs in the gene body and upstream and downstream 5 kb regions of the three genes presented substantially different allele frequencies between small-sized and large-sized pigs.

Selection and functional exploration of the *GALNT13* gene

For *GALNT13*, the rs13030825 allele is associated with increased humeral length and height in African Americans⁴⁰. A linkage disequilibrium analysis of the 5-kb flanking regions of upstream and downstream of *GALNT13* (chr15: 69,791–70,392 Mb) revealed that haplotypes are highly differentiated between small and large pigs, including one covering a 4.3-kb PUR within the fourth intron of *GALNT13* (Supplementary Fig. 17a). Given the strong selection signals, the *GALNT13* gene was selected for further exploration of its function in regulating body length. The evolution of vertebrate body length has been largely driven by modifications in the axial skeleton, and endochondral ossification is an intrinsic driver of trunk and limb bone formation⁴¹. We found that *GALNT13* is highly expressed in pig bone-related tissues, including thoracic vertebral cartilage, lumbar cartilage, epiphyseal cartilage and bone marrow (Fig. 6a). Immunohistochemical (IHC) staining revealed high expression of *GALNT13* in thoracic vertebral cartilage (Fig. 6b), particularly in hypertrophic chondrocytes. Moreover, substantially lower expression levels of the *GALNT13* gene were detected in the thoracic vertebral cartilage of Wuzhishan pigs at 3 days, 30 days, 60 days and 120 days after birth (Fig. 6c), when compared to that of Yorkshire pigs.

We successfully isolated thoracic vertebral chondrocytes and epiphyseal chondrocytes (Supplementary Fig. 17b), as confirmed by the immunofluorescence detection of COL2A1, an early and abundant marker of chondrocytes. The in vitro differentiation induction revealed that the expression of *GALNT13* continued to increase from 1 day to 7 days, in both thoracic vertebral chondrocytes and epiphyseal chondrocytes (Supplementary Fig. 17c). Subsequently, porcine primary chondrocytes were transfected with a *GALNT13* overexpression plasmid using Lipofectamine 2000 during culture in differentiation medium. *GALNT13* overexpression in thoracic vertebral chondrocytes and epiphyseal chondrocytes promoted the expression of the chondrocyte marker genes *COL2A1*, *SOX9* and *RUNX2*, indicating enhanced chondrocyte differentiation (Fig. 6d,e). In addition, a cell proliferation assay revealed that *GALNT13* overexpression inhibited the proliferation of porcine primary chondrocytes (Fig. 6f–i). In summary, our results indicate that the dynamic expression of *GALNT13* has a role in chondrocyte proliferation and differentiation, potentially contributing to body size in pigs.

Discussion

In this study, we present the complete T2T gap-free pig genome assembly T2T-pig1.0 with >10% unknown sequences (that is, PURs). More than 90% PURs are located in genomic regions that are inherently difficult to assemble and are typically dominated by repetitive elements, including the highly repetitive sequences and multigene families (for example, olfactory receptor genes). Given the 1–2 million years of evolutionary divergence between Asian pig lineages and European/North American breeds, the PURs also include some lineage-specific sequences that are absent from European domestic breeds, such as those found in the Duroc for the Sscrofa11.1 assembly. T2T-pig1.0 enabled the precise characterization of centromeric repeats and the chrY, and uncovered new genes and SVs, including a ~30-Mb inversion (chr10–SV1).

The 11 YOGs conserved in MSY across complete chrY T2T assemblies of humans, sheep, goats and pigs, including *DDX3Y*, *USP9Y*, *ZFY*, *HSFY*, *TSPY* and *SRY*, represent core genetic elements essential for spermatogenesis and testicular development^{42–47}. Multicopy YOGs on chrY are frequently observed across diverse animal species. For example, 8 *HSFY* copies have been reported in humans²⁴, 14 *HSFY* copies in T2T-pig1.0, 20–145 *HSFY* copies in buffalo⁴⁸ and 1–3 copies in Bactrian camel⁴⁹. The *HSFY* copy number is correlated with the sire conception rate in cows⁵⁰. The *HSFY* family in domestic pigs and other suid species has undergone expansion, with multiple copies located on the chrY⁵¹. We revealed variations in *HSFY* copy numbers on chrY across different pig genome assemblies, with 14 and 25 copies in T2T-pig1.0 and Sscrofa11.1, respectively. This variation in the *HSFY* copy number may be associated with the superior male reproductive performance observed in Duroc pigs, which are frequently selected as terminal sires in commercial crossbreeding programs. The *SRY* gene initiates the development of testes from the bipotential cells of the early gonad. Consistent with the findings of previous studies^{23,52}, two *SRY* genes in pigs were confirmed to increase the superior reproductive ability of males, whereas only one copy was reported in the T2T chrY assemblies of humans, sheep and goats^{10,24,53}. *SRY* copy numbers vary across the populations of horses⁵⁴ (1–2.7 copies), swamp buffaloes⁵⁵ (1–4 copies), dogs⁵⁶ (1–3 copies) and rats⁵⁷ (1–11 copies). The other five YOGs (*KDM6A*, *AMELY*, *OFD1*, *BCOR* and *EIF1AX*) in MSY were found with the paralogous genes on chrX and autosomes. The high variability in the copy number of YOGs in MSY is likely caused by the instability and degeneration of the chrY due to the lack of recombination⁵⁸.

In conclusion, T2T-pig1.0 represents a comprehensive and complete genomic resource for agricultural practice and biomedical research on pigs. An obvious topic for future studies involves the in-depth functional characterization of genes in the PURs. Given the critical role of centromeres in chromosomal segregation, identifying population variations in centromeric regions in wild boars and

domestic pigs will be highly beneficial for studying centromere evolution events during pig domestication.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41588-025-02433-6>.

References

- Larson, G. et al. Worldwide phylogeography of wild boar reveals multiple centers of pig domestication. *Science* **307**, 1618–1621 (2005).
- Groenen, M. A. et al. Analyses of pig genomes provide insight into porcine demography and evolution. *Nature* **491**, 393–398 (2012).
- Wang, X. et al. Genetic characteristics of inbred Wuzhishan miniature pigs, a native Chinese breed. *J. Reprod. Dev.* **52**, 639–643 (2006).
- Niu, M. et al. Long-term case study of a Wuzhishan miniature pig with diabetes. *Animal Model. Exp. Med.* **3**, 22–31 (2020).
- Guo, Q. et al. Production of cloned Wuzhishan miniature pigs and application for alloxan toxicity test. *Anim. Biotechnol.* **26**, 292–297 (2015).
- Fang, Y. et al. An animal model study on the gene expression profile of meniscal degeneration. *Sci. Rep.* **10**, 21469 (2020).
- Warr, A. et al. An improved pig reference genome sequence to enable pig genetics and genomics research. *GigaScience* **9**, g1aa051 (2020).
- Nurk, S. et al. The complete sequence of a human genome. *Science* **376**, 44–53 (2022).
- Luo, L. Y. et al. Telomere-to-telomere sheep genome assembly identifies variants associated with wool fineness. *Nat. Genet.* **57**, 218–230 (2025).
- Wu, H. et al. Telomere-to-telomere genome assembly of a male goat reveals variants associated with cashmere traits. *Nat. Commun.* **15**, 10041 (2024).
- Liu, J. et al. The complete telomere-to-telomere sequence of a mouse genome. *Science* **386**, 1141–1146 (2024).
- Zong, W. et al. Two telomere-to-telomere pig genome assemblies and pan-genome analyses provide insights into genomic structural landscape and genetic adaptations. *iMeta* **4**, e70013 (2025).
- Swanson, K. S. et al. Genomics and clinical medicine: rationale for creating and effectively evaluating animal models. *Exp. Biol. Med.* **229**, 866–875 (2004).
- Hu, J. et al. NextDenovo: an efficient error correction and accurate assembly tool for noisy long reads. *Genome Biol.* **25**, 107 (2024).
- Cheng, H., Concepcion, G. T., Feng, X., Zhang, H. & Li, H. Haplotype-resolved de novo assembly using phased assembly graphs with hifiasm. *Nat. Methods* **18**, 170–175 (2021).
- Manni, M., Berkeley, M. R., Seppey, M., Simao, F. A. & Zdobnov, E. M. BUSCO update: novel and streamlined workflows along with broader and deeper phylogenetic coverage for scoring of eukaryotic, prokaryotic, and viral genomes. *Mol. Biol. Evol.* **38**, 4647–4654 (2021).
- Mellink, C. H., Bosma, A. A. & de Haan, N. A. Variation in size of Ag-NORs and fluorescent rDNA in situ hybridization signals in six breeds of domestic pig. *Hereditas* **120**, 141–149 (1994).
- Gustavsson, I. Standard karyotype of the domestic pig. Committee for the standardized karyotype of the domestic pig. *Hereditas* **109**, 151–157 (1988).
- Quenet, D. & Dalal, Y. The CENP-A nucleosome: a dynamic structure and role at the centromere. *Chromosome Res.* **20**, 465–479 (2012).
- Logsdon, G. A. et al. The structure, function and evolution of a complete human chromosome 8. *Nature* **593**, 101–107 (2021).
- Altmeose, N. et al. Complete genomic and epigenetic maps of human centromeres. *Science* **376**, eabl4178 (2022).
- Chang, C. H. et al. Islands of retroelements are major components of *Drosophila* centromeres. *PLoS Biol.* **17**, e3000241 (2019).
- Skinner, B. M. et al. The pig X and Y chromosomes: structure, sequence, and evolution. *Genome Res.* **26**, 130–139 (2016).
- Rhie, A. et al. The complete sequence of a human Y chromosome. *Nature* **621**, 344–354 (2023).
- McGraw, R. A., Jacobson, R. J. & Akamatsu, M. A male-specific repeated DNA sequence in the domestic pig. *Nucleic Acids Res.* **16**, 10389 (1988).
- Zech, D., Rana, S., Buchler, M. W. & Zoller, M. Tumor-exosomes and leukocyte activation: an ambivalent crosstalk. *Cell Commun. Signal.* **10**, 37 (2012).
- He, Y. et al. P2RY8 variants in lupus patients uncover a role for the receptor in immunological tolerance. *J. Exp. Med.* **219**, e20211004 (2022).
- Goswami, D. et al. Endothelial CD99 supports arrest of mouse neutrophils in venules and binds to neutrophil PILRs. *Blood* **129**, 1811–1822 (2017).
- Marchini, A., Ogata, T. & Rappold, G. A. A track record on SHOX: from basic research to complex models and therapy. *Endocr. Rev.* **37**, 417–448 (2016).
- Wang, Y. et al. A chromosome-level genome of Chenghua pig provides new insights into the domestication and local adaptation of pigs. *Int. J. Biol. Macromol.* **270**, 131796 (2024).
- Huang, M. et al. The fine-scale genetic structure and selection signals of Chinese indigenous pigs. *Evol. Appl.* **13**, 458–475 (2020).
- Ai, H. et al. Adaptation and possible ancient interspecies introgression in pigs identified by whole-genome sequencing. *Nat. Genet.* **47**, 217–225 (2015).
- Rubin, C. J. et al. Strong signatures of selection in the domestic pig genome. *Proc. Natl Acad. Sci. USA* **109**, 19529–19536 (2012).
- Mikawa, S. et al. Fine mapping of a swine quantitative trait locus for number of vertebrae and analysis of an orphan nuclear receptor, germ cell nuclear factor (NR6A1). *Genome Res.* **17**, 586–593 (2007).
- Xia, T. et al. Genomic insights into the mechanisms of body size evolution in Serpentes. *BMC Genomics* **26**, 420 (2025).
- Trilaksono, B. et al. Body weight and measurement traits of Brahman cattle affected by pleomorphic adenoma gene 1 variations. *Open Vet. J.* **14**, 3248–3256 (2024).
- Signer-Hasler, H., Burren, A., Ammann, P., Drogemüller, C. & Flury, C. Runs of homozygosity and signatures of selection: a comparison among eight local Swiss sheep breeds. *Anim. Genet.* **50**, 512–525 (2019).
- Jee, Y. H., Baron, J. & Nilsson, O. New developments in the genetic diagnosis of short stature. *Curr. Opin. Pediatr.* **30**, 541–547 (2018).
- Nordberg, R. C., Wang, H., Wu, Q. & Lobo, E. G. Corin is a key regulator of endochondral ossification and bone development via modulation of vascular endothelial growth factor A expression. *J. Tissue Eng. Regen. Med.* **12**, 2277–2286 (2018).
- Tekola-Ayele, F. et al. Admixture mapping identifies African and Amerindigenous local ancestry loci associated with fetal growth. *Hum. Genet.* **140**, 985–997 (2021).
- Mackie, E. J., Ahmed, Y. A., Tatarczuch, L., Chen, K. S. & Mirams, M. Endochondral ossification: how cartilage is converted into bone in the developing skeleton. *Int. J. Biochem. Cell Biol.* **40**, 46–62 (2008).
- Krausz, C., Giachini, C. & Forti, G. TSPY and male fertility. *Genes* **1**, 308–316 (2010).

43. Harley, V. R., Clarkson, M. J. & Argentaro, A. The molecular action and regulation of the testis-determining factors, SRY (sex-determining region on the Y chromosome) and SOX9 [SRY-related high-mobility group (HMG) box 9]. *Endocr. Rev.* **24**, 466–487 (2003).
 44. Ahmadi Rastegar, D. et al. Isoform-level gene expression profiles of human Y chromosome azoospermia factor genes and their X chromosome paralogs in the testicular tissue of non-obstructive azoospermia patients. *J. Proteome Res.* **14**, 3595–3605 (2015).
 45. Holmlund, H., Yamauchi, Y., Ruthig, V. A., Cocquet, J. & Ward, M. A. Return of the forgotten hero: the role of Y chromosome-encoded Zfy in male reproduction. *Mol. Hum. Reprod.* **29**, gaad025 (2023).
 46. Sun, C. et al. An azoospermic man with a de novo point mutation in the Y-chromosomal gene USP9Y. *Nat. Genet.* **23**, 429–432 (1999).
 47. Vakilian, H. et al. DDX3Y, a male-specific region of Y chromosome gene, may modulate neuronal differentiation. *J. Proteome Res.* **14**, 3474–3483 (2015).
 48. Zhang, X. et al. HSFY and ZNF280BY show copy number variations within 17 water buffalo populations. *Anim. Genet.* **48**, 221–224 (2017).
 49. Chen, H., Ren, Z., Zhao, J., Zhang, C. & Yang, X. Y-chromosome polymorphisms of the domestic Bactrian camel in China. *J. Genet.* **97**, 3–10 (2018).
 50. Yue, X. P. et al. Copy number variations of the extensively amplified Y-linked genes, HSFY and ZNF280BY, in cattle and their association with male reproductive traits in Holstein bulls. *BMC Genomics* **15**, 113 (2014).
 51. Skinner, B. M. et al. Expansion of the HSFY gene family in pig lineages: HSFY expansion in suids. *BMC Genomics* **16**, 442 (2015).
 52. Tao, D. et al. A simple, affordable, and rapid visual CRISPR-based field test for sex determination of earlier porcine embryos and pork products. *Mol. Biotechnol.* **65**, 263–272 (2023).
 53. Olagunju, T. A. et al. Telomere-to-telomere assemblies of cattle and sheep Y-chromosomes uncover divergent structure and gene content. *Nat. Commun.* **15**, 8277 (2024).
 54. Castaneda, C. et al. Copy number variation of horse Y chromosome genes in normal equine populations and in horses with abnormal sex development and subfertility: relationship of copy number variations with Y haplogroups. *G3* **12**, jkac278 (2022).
 55. Sun, T., Hanif, Q., Chen, H., Lei, C. & Dang, R. Copy number variations of four Y-linked genes in swamp buffaloes. *Animals* **10**, 31 (2019).
 56. Krzeminska, P., Nowacka-Woszek, J. & Switonski, M. Copy number variation of the SRY gene showed an association with disorders of sex development in Yorkshire Terrier dogs. *Anim. Genet.* **53**, 152–155 (2022).
 57. Prokop, J. W. et al. Analysis of Sry duplications on the Rattus norvegicus Y-chromosome. *BMC Genomics* **14**, 792 (2013).
 58. Lenormand, T. & Roze, D. Y recombination arrest and degeneration in the absence of sexual dimorphism. *Science* **375**, 663–666 (2022).
 59. Jenkins, T. L. MAPMIXTURE: an R package and web app for spatial visualisation of admixture and population structure. *Mol. Ecol. Resour.* **24**, e13943 (2024).
- Publisher's note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.
- Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.
- © The Author(s), under exclusive licence to Springer Nature America, Inc. 2025

Methods

Sample collection and sequencing

Fresh whole blood and tissue samples were collected from a 120-day-old healthy boar from the Wuzhishan Pig Breeding Farm of the Hainan Academy of Agricultural Sciences. For the population analysis, 30 Ding'an pigs and 20 Wuzhishan pigs were also sampled from the Wuzhishan Pig Breeding Farm using the same methods. Specifically, fresh blood was collected with a 20-ml syringe and immediately transferred to a K2 EDTA blood collection tube (Xinle, V272739), before being brought back to the laboratory on ice. All the tissue samples were collected immediately after slaughter, frozen in liquid nitrogen and stored at -80°C . All animal procedures were performed in accordance with the guidelines for animal experiments approved by the China Agricultural University Institutional Animal Care and Use Committee (AW31904202-1-1). DNA and RNA extraction, library construction, and MGI, PacBio HiFi, ONT, Bionano and Hi-C sequencing were performed at GrandOmics (Supplementary Methods).

Draft assembly

The initial assemblies for T2T-pig1.0 were created using the default configuration for Hifiasm (v0.19.4) and NextDenovo (v2.5.2), which were run based on HiFi and ONT reads, respectively. The assemblies were evaluated based on the number and N50 of both contigs and scaffolds, and the contigs obtained by Hifiasm, rather than the contigs obtained by NextDenovo, were used for the subsequent assembly and polishing. Contamination of bacterial sequences and non-nuclear mitochondrial sequences were removed by alignment against the NCBI nucleotide NR database using BLASTN (v2.10.0)⁶⁰ with a '-evalue' parameter of 1×10^{-5} . The two haplotypes were assembled based on the ONT, HiFi and Hi-C reads, by using Hifiasm with the parameter of '--dual-scaf'.

ChrY assembly

For the independent chrY assembly (Supplementary Fig. 1), HiFi and ONT reads polished by NextDenovo were used to assemble initial contigs using Hifiasm with the parameter '--ul --dual-scaf', with the assistance of Hi-C data. All the scaffolds were subsequently aligned with the Sscrofa11.1 genome to identify the scaffolds originating from the chrY. Only one scaffold (~43 Mb) with one gap, which is estimated to be 1.3 Mb, was selected, and this scaffold was further checked and adjusted based on the Hi-C interaction signals. Gap filling, telomere patching and polishing were performed together with the other chromosomes. Finally, we produced a complete T2T chrY without any gaps.

Pseudochromosome construction

The raw Hi-C reads were subjected to quality control using the fastp software (v0.23.4)⁶¹ with the parameters '-f 6 -F 6 -t 4 -T 4' to generate high-quality clean reads. Clean reads were aligned to the contigs of autosomes and chrX and chrY using Bowtie2 (v2.4.2)⁶². Scaffold clustering was executed for chromosome placement using LACHESIS⁶³ with the parameters 'CLUSTER MIN RE SITES = 100, CLUSTER MAX LINK DENSITY = 2.5, CLUSTER NONINFORMATIVE RATIO = 1.4, ORDER MIN N RES IN TRUNK = 60, and ORDER MIN N RES IN SHREDS = 60'. Any placement and orientation inaccuracies were manually corrected and, based on the alignment against Sscrofa11.1, 20 pseudochromosomes in T2T-pig1.0 and the 2 haplotypes, including 18 autosomes and chrX and chrY, were determined for their orientation and ordering.

Gap filling

To close the gaps, ultralong ONT reads exceeding 100 kb in length were mapped to the 20 pseudochromosomes using Minimap2 (v2.23)⁶⁴ with the '-x map-ont' option. Ultralong ONT reads that were accurately aligned to the adjacent ends of both contigs in a gap (identity, $\geq 95\%$; coverage, $\geq 90\%$; and QV, ≥ 20) were chosen as anchor sequences. Short gaps were filled by simply extending these anchor sequences. For larger gaps, three types of ultralong ONT reads were selected based on their

alignment results for local assembly. In addition to the first type of anchor sequences, the second type of ONT reads, which were roughly aligned to the anchor sequences (identity, $< 95\%$, and coverage, $< 90\%$), were also retrieved. The third type consisted of reads that did not align with any pseudochromosomes of the whole assembly, some of which might have originated from a specific gap. The aforementioned three types of reads were then aligned to each other using Minimap2 (v2.23) to generate a library facilitating the overlapping of pairwise ONT reads. The *k*-mer hash tables were computed from MGI short reads using Jellyfish (v2.3.0)⁶⁵, and rare or low-frequency *k*-mers were identified when their depth was below the average. A string graph for a specific gap was subsequently constructed using the first type of read as starting and ending points with the NextGraph module of NextDenovo (v2.5.2), based on the overlapping ONT reads and their rare *k*-mers. The graph with the longest cumulative length of rare *k*-mers was selected to finalize the sequences to close the gaps. Finally, all ONT reads were mapped to the gap-filled genome using Integrative Genomics Viewer (IGV; v2.13)⁶⁶, with manual adjustments made through multiple iterations to assess the coverage and depth of the gap regions.

Genome polishing

Based on the alignments of the HiFi reads using Minimap2 (v2.23), low-quality regions were identified using the following criteria: MAPQ score of ≤ 1 , clipped reads at both ends and fewer than three aligned HiFi reads. High-quality and low-quality regions were subjected to separate polishing processes. The low-quality regions were polished sequentially based on the long HiFi reads using Nextpolish2 (v0.1.0)⁶⁷, as well as ONT long reads and MGI short reads using Nextpolish (v1.4.1)⁶⁸. The high-quality regions were subjected to additional polishing based on the long HiFi reads using Nextpolish2 before being combined into a complete genome with polished low-quality regions. Another round of polishing was performed based on the HiFi reads using Nextpolish2 to prevent errors during the merging process and to create the final seamless genome assembly, T2T-pig1.0.

Telomere assembly

We combined three categories of HiFi reads for independent telomere assembly—those with more than ten telomere-specific repeats (that is, AACCT or AGGGTT), those unmapped to T2T-pig1.0 and those that were only partially aligned to the 1-Mb chromosomal incomplete ends. Telomere assembly was conducted using Hifiasm (v0.19.4) with the default settings. The contigs containing telomeric sequences were aligned to T2T-pig1.0 using Minimap2 (v2.23) to identify their potential chromosomal locations. Telomeres were positioned at the 1 Mb ends of each chromosome by leveraging the overlap between contigs and chromosomal incomplete ends using RagTag (v2.1.0)⁶⁹. The updated chromosomal ends were manually inspected to ensure complete alignment of the HiFi and ONT long reads.

Validation of the T2T genome assembly

MGI short reads, along with ONT and HiFi long reads, were aligned to the T2T-pig1.0 genome using BWA-MEM (v0.7.17-r1188)⁷⁰ and Minimap2 (v2.23), respectively. The genome coverage was then assessed with a 1-Mb window size using Bamdst (<https://github.com/shiquan/bamdst>), and visualized with the KaryoPloteR package (v1.8.4)⁷¹. For the collinearity analysis shown in Extended Data Figs. 4 and 8a, a comparison between the two chromosomes was performed using Minimap2 (v2.23), and genome synteny was depicted using paf2dotplot (<https://github.com/moold/paf2dotplot>) and GenomeSyn⁷². The completeness of the T2T-pig1.0 assembly was assessed using BUSCO (v5.1.3)⁷³, with the OrthoDB10 mammalian database used as a reference. QVs were calculated for the entire genome assembly using Merqury (v1.3)⁷⁴ to evaluate the consensus accuracy (Supplementary Fig. 3). Based on the alignment of T2T-pig1.0 and Sscrofa11.1 using Minimap2, the chromosomal regions in T2T-pig1.0 that were not found in the chromosomes

and unplaced contigs of Sscrofa11.1 were determined to be PURs in T2T-pig1.0 by using BEDTools (v2.30.0)⁷⁵.

Annotation of repeat sequences

The annotation of repeat sequences, including tandem repeats, transposons and satellites, was performed using both de novo and homology-based prediction methods. The TE libraries were generated by training with EDTA (v2.2)⁷⁶, and then all the repeat elements were masked by RepeatMasker (v4.1.4)⁷⁷ with the default settings. SDs were identified with BISER (v1.4)⁷⁸, and the low-quality SDs were filtered based on the following criteria: (1) sequence identity of >90%, (2) ≤50% gaps in the alignment, (3) aligned sequence length of ≥1 kb and (4) satellite sequences of ≤70% as annotated by RepeatMasker.

Gene annotation

The protein-coding genes in the T2T-pig1.0 assembly were annotated using the following three approaches: de novo, homology-based and transcriptome-based approaches (Supplementary Fig. 7a). For the transcriptome-based approach, the 15 RNA-seq datasets from pig tissues (Supplementary Table 1) were first aligned using STAR (v2.7.9a), and then the transcripts were constructed using StringTie (v2.2.1)⁷⁹, with the parameters ‘-j 2 -f 0.01 -c 2 -m 200 -a 10’. The full-length transcripts from Iso-seq were aligned to T2T-pig1.0 using GMAP (<https://github.com/juliangehring/GMAP-GSNAP>). For homology-based predictions, protein sequences from five closely related species (pig, NCBI accession [GCA_000003025.6](#); bovine, GCF_002263795.3; human, GCF_009914755.1; rat, [GCA_036323735.1](#) and mouse, [GCA_000001635.9](#)) were retrieved from the NCBI database. Gene models were predicted through alignment of these sequences against T2T-pig1.0 using GeMoMa (v1.9)⁸⁰ with the parameters ‘Annotation-Finalizer.r=NO Extractor.r=true Extractor.s=flase DenoiseIntrons.c=15 GeMoMa.m=100000’. In the de novo prediction approach, gene predictions were made using AUGUSTUS (v3.4.0)⁸¹ with the parameters ‘--UTR=off --uniqueGenId=true --noInFrameStop=true --gff3=on --strand=both --genemodel=complete’. All gene model predictions were integrated using MAKER (v3.01.03 (ref. 82); Supplementary Fig. 7a).

The software Infernal (v1.1.4)⁸³ was used to predict genomic ncRNAs (rRNAs, miRNAs, snRNAs and tRNAs) based on the Rfam database (v.14.9)⁸⁴, with the parameters ‘infernal-1.1.4 --rfam --nohmmonly --cut_ga’. The programs tRNAscan-SE (v2.0.12)⁸⁵ and RNAmme (v1.2)⁸⁶ were used to identify tRNAs and rRNAs, respectively. The above ncRNAs were identified and merged through an integrated bioinformatic pipeline (Supplementary Fig. 7b). The annotation of pseudogenes and lncRNAs in T2T-pig1.0 was performed using the NCBI eukaryotic genome annotation pipeline (https://www.ncbi.nlm.nih.gov/refseq/annotation_euk/process/).

Functional annotation of protein-coding genes

Protein sequences were annotated against three databases, namely, Swiss-Prot⁸⁷ (http://web.expasy.org/docs/swiss-prot_guideline.html), the Kyoto Encyclopedia of Genes and Genomes⁸⁸ (KEGG, <https://www.genome.jp/kegg/>), and the NCBI NR protein database (<ftp://ftp.ncbi.nih.gov/blast/db/>), using BLASTP (v2.10.0)⁶⁰ with the parameters ‘-evalue 1e-5 -max_target_seqs 1’. In addition, proteins were matched to the Pfam database⁸⁹ (<https://pfam-legacy.xfam.org/>) using InterProScan (v5.58)⁹⁰, and GO terms (<http://geneontology.org>) were extracted.

Centromere identification

For the ChIP assay, 10 ml of fresh blood from the same boar used to construct the T2T assembly was treated with 1% formaldehyde for 15 min at room temperature to cross-link the proteins. This reaction was then halted by adding glycine at a final concentration of 200 mmol l⁻¹. Sonication was subsequently performed to produce soluble sheared chromatin DNA with an average length of 200–500 bp. Next, immunoprecipitation was performed using a monoclonal antibody against

CENP-A (ADI-KAM-CC006-E, Enzo Life Sciences). The DNA from the immunoprecipitation reaction was subsequently used to construct sequencing libraries with the NEXTflex ChIP-seq Library Prep Kit for Illumina sequencing (Bioo Scientific, NOVA-5143-01). These libraries were sequenced on an Illumina NovaSeq 6000 platform in paired-end mode (Illumina). After trimming with fastp (v0.23.4), the clean reads were aligned to T2T-pig1.0 using Bowtie2 (v2.4.2) with the parameters ‘--very-sensitive --no-mixed --no-discordant -k 10’. The read depth for ChIP enrichment in 5-kb windows was determined by using BEDTools (v2.30.0) and visualized with the KaryoPloteR package (v1.8.4). Heatmaps of sequence identity for centromeric repeats were created using StainedGlas (v0.5)⁹¹, with the color coding representing the percentage identity.

Identification of repetitive sequences inside centromeres

Initial centromeric regions were identified using ChIP-seq signals. A *k*-mer library was constructed from the identified centromere regions using KMC (v3.1.1)⁹² with the parameters ‘-fm -k151 -ci20 -cs100000’. Centromeric repeat units were then detected based on the *k*-mer frequencies using the SRF program, and these units were grouped into three categories of satellite repeat units (SAT1, SAT2 and SAT3) according to their sequence identities. The centromeric satellite repeats were subsequently realigned to T2T-pig1.0 to assess their distribution using BLASTN (v2.10.0). The final centromeric regions were defined by analyzing the distribution patterns of these three types of repeat units.

Methylation determination based on long reads

DNA methylation was analyzed through HiFi reads. HiFi reads were aligned to T2T-pig1.0 using pbmm2 (v1.13.0, <https://github.com/PacificBiosciences/pbmm2>) with default settings. The resulting BAM file was processed using the ‘aligned_bam_to_cpg_scores’ command from pb-CpG-tools (v2.3.2, <https://github.com/PacificBiosciences/pb-CpG-tools>) to determine the site probabilities of 5-methylcytosine methylation, which were then visualized using the IGV program (v2.13). Methylation coverage was evaluated across 50-kb windows using BEDTools (v2.30.0).

Estimation and validation of rDNA copy number

The total rDNA copy number was estimated using MGI, HiFi and ultra-long ONT reads, according to the previously established methods^{8,93}. The ddPCR was performed using the QX100 Droplet Digital PCR System platform (Bio-Rad) to validate the 45S rDNA copy number using the following primers: forward–5′-AACGTGAGCTGGGTTAG-3′ and reverse–5′-CTCGTACTGAGCAGGATTAC-3′. The single-copy gene *GCG* was used to calculate the ddPCR copy number values, with the following primers: forward–5′-GAATCAACACCATCGGTCAAAT-3′ and reverse–5′-CTCCACCCATAGAATGCCAGT-3′.

LTR insertion

To calculate the LTR insertion time across the T2T-pig1.0, human, sheep and goat genome assemblies, we followed the ‘Ageing LTR Insertions’ pipeline^{94,95} (https://github.com/SIWLab/Lab_Info/wiki/Ageing-LTR-insertions). In brief, intact LTR retrotransposons were identified using LTRharvest⁹⁶, a program that scans the genome for LTR sequences based on specific motifs and the characteristic structure of LTR retrotransposons. Each pair of 5′ and 3′ LTRs was realigned to each other and saved in a GFF3 file. The sequences of every pair of LTRs were further aligned using Muscle (v3.8.31)⁹⁷. With the gaps trimmed, the distance between two LTRs was estimated based on the Kimura two-parameter model, and the insertion time was calculated based on the substitution rates of 1.2×10^{-9} , 4.1×10^{-10} , 2.16×10^{-9} and 8.3×10^{-9} substitutions per site per year for pigs⁹⁸, humans⁹⁸, goats⁹⁹ and sheep¹⁰⁰, respectively. The LTR insertion time data were visualized using TBtools (v2.056)¹⁰¹, and comparisons between the centromeric and noncentromeric regions were performed by using the R package ‘ggpubr’.

Defining the PAR boundary

We determined the PAR boundary on the chrY of T2T-pig1.0, as previously described^{53,102}. Briefly, all the HiFi reads were aligned to the chrY and visualized using IGV (v2.13). The alignment track (Extended Data Fig. 8b) distinctly displayed the PAR boundary, where soft clipping of the long reads occurred due to no similarity of maternal HiFi reads with the chrY. A sharp reduction in read coverage was observed beyond the PAR boundary, dropping to almost half of that in the PAR, as shown in the coverage track (Extended Data Fig. 8b).

Cross-species comparative analysis

The genomes of six species—humans, mice, horses, wolves, cattle and pigs—were selected for the evolutionary analysis. Gene families were determined by using OrthoFinder (v2.5.3, <https://github.com/davidemms/OrthoFinder>) based on the complete amino acid sequences. Functional enrichment analyses, including GO and KEGG pathway enrichment analyses, were performed using clusterProfiler (v3.19)¹⁰³. Multiple sequence alignment was performed based on the protein sequences of single-copy gene families. A phylogenetic tree of the species was constructed using the RAXML (v1.2.2)¹⁰⁴ with the following parameters ‘-m GTRGAMMA -p 12345 -f a -o DROME -x 12345 -# 100’. Divergence time was estimated using the MCMCtree program in the PAML¹⁰⁵ package. The expansion and contraction of gene families in each species were predicted using CAFE (v1.1)¹⁰⁶. The macro-synteny relationships for the whole genome, chrY and PARs among the available T2T genome assemblies of humans, sheep, goats and pigs were analyzed using JCVI (v1.4.25)¹⁰⁷.

SNP calling based on short reads

Whole-genome sequencing data from 407 domesticated and wild pig individuals were used for SNP calling, including 50 individuals from Wuzhishan and Ding'an pigs sequenced in this study and the other 357 individuals obtained from public databases (Supplementary Table 12). Quality control of the raw sequences was conducted using FastQC (v0.11.9, <https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>) and Trimmomatic (v0.39)¹⁰⁸. The cleaned short reads were aligned to both T2T-pig1.0 and Sscrofa11.1 using BWA-MEM (v0.7.17-r1188)¹⁰⁹ with the parameters ‘-k 23 -M’. The BAM files were then sorted with SAMtools (v1.16)¹¹⁰ and subsequently used for SNP calling using GATK (v4.3)¹¹¹. Raw SNPs were filtered through the VariantFiltration module of GATK with the parameters ‘QD < 2.0 || QUAL < 30.0 || SOR > 3.0 || FS > 60.0 || MQ < 40.0 || MQRankSum < -12.5 || ReadPosRankSum < -8.0’. After SNPs with more than 10% missing genotypes and a minor allele frequency of less than 0.05 were eliminated using VCFtools (v0.1.16)¹¹⁰, high-quality biallelic SNPs were retained for subsequent analyses.

Population genetic analyses

A phylogenetic tree was generated based on the *p*-distance data derived from high-quality autosomal biallelic SNPs using the neighbor-joining method implemented in PHYLIP (v3.697, <https://phylipweb.github.io/phylip/>) and subsequently visualized using iTOL (v6.8.1)¹¹². Principal component analysis was conducted on these independent SNPs using the ‘smartpca’ function within the EIGENSOFT package (v8.0.0)¹¹³. Additionally, the population genetic structure was assessed using ADMIXTURE (v1.3.0)¹¹⁴ based on the independent SNPs.

Selective sweeps for body size

To detect selective signals based on SNPs, F_{ST} values were computed using VCFtools (v0.1.16)¹¹⁰ across 50-kb windows with a 10-kb step size between large-sized and small-sized pigs. Nucleotide diversity (π) was assessed within 100-kb windows using VCFtools. The signatures with the top 5% of the F_{ST} scores were regarded as candidate selective sweeps potentially linked to body stature, with π ratios between pairwise populations used to validate these selective sweeps. ANNOVAR¹¹⁵ software

was used to annotate all the selected SNPs with extreme differences in allele frequency between large-sized and small-sized pigs (>80% in one group and <20% in the other).

Permutation test for overlapping selected regions

We examined overlaps among selective sweep signals across four comparative analyses (two large-sized versus two small-sized pig groups). We assessed the statistical significance by performing permutation tests to determine whether these observed overlaps differed substantially from random expectations. Simulated datasets for each of the four comparison groups were generated by randomly distributing the genomic regions that matched both the quantity and size distribution of the observed selective sweep regions, using the shuffle command of BEDTools (v2.30.0). The common regions were obtained across the four comparative analyses. This randomization process was repeated 10,000 times to establish empirical null distributions. We compared the observed overlap count against these null distributions, calculating empirical *P* values as the proportion of permutations where simulated overlaps equaled or exceeded the observed count.

Analysis of *GALNT13* expression

Amplification was performed by qRT-PCR with the primer pairs for the *GALNT13* gene (Supplementary Table 15). The cycling parameters were as follows: initial heat denaturation at 95 °C for 5 min; 28 cycles of 95 °C for 30 s, 60 °C for 30 s and 72 °C for 30 s; and a final extension at 72 °C for 5 min. The PCR products were analyzed by electrophoresis on a 2% agarose gel and were subsequently photographed.

IHC staining for GALNT13 protein was performed. Briefly, sections were prepared from paraffin-embedded cartilage tissue blocks and stained with a rabbit polyclonal primary antibody against GALNT13 (Boster Bio, PB0675) using immunoperoxidase-based IHC. Goat antirabbit secondary antibody labeled with horseradish peroxidase (ServiceBio, GB23303) was used for secondary detection. After IHC staining, the sections were counterstained with hematoxylin solution, and the images were acquired using SlideView software (v2.8.0).

Cell isolation and culture

Primary chondrocytes were isolated from the thoracic intervertebral cartilage and epiphyseal cartilage of newborn piglets. The tissues were minced with surgical scissors and digested with a freshly prepared collagenase D solution¹¹⁶. The isolated cells were subsequently characterized by immunofluorescence staining with a COL2A1 antibody (Boster Bio, PA2141-1). All the cells were cultured in DMEM supplemented with 10% FBS and 1% penicillin–streptomycin. For chondrocyte differentiation, porcine primary chondrocytes were cultured in a medium enriched with 1% insulin–transferrin–selenium (Gibco, 41400045). For transfection, porcine primary chondrocytes were transfected in a six-well plate with 2 µg of overexpression vector or empty plasmid (Supplementary Table 15) and Lipofectamine 2000 (Invitrogen, 11668019)¹¹⁷. The cells were collected for RNA extraction 2 days after transfection.

Cell differentiation and proliferation assays

After chondrocyte differentiation was induced, total RNA was extracted with TRIzol Reagent (Invitrogen, 15596018CN) according to the manufacturer's protocol. Reverse transcription was performed with the FastKing RT Kit with gDNase (TIANGEN Biotech, KR116-02) according to the manufacturer's instructions. qRT-PCR was conducted on a QuantStudio 7 Flex real-time PCR system (Applied Biosystems) using SuperReal PreMix Plus (SYBR Green; TIANGEN Biotech, FP205-02). Gene expression levels were normalized with porcine β -actin as a control, and the primers used are shown in Supplementary Table 15. Chondrocyte proliferation was detected using a CCK-8 kit (Beyotime Biotechnology, C0039) and a BeyoClick EdU-488 kit (Beyotime Biotechnology, C0071L) according to the manufacturer's instructions.

Statistics and reproducibility

All the statistical analyses were performed using GraphPad Prism (v.9.5, <https://www.graphpad.com>) or R (v.4.4.2). Detailed information on all the statistical analyses, including the tests and their sample sizes (n), is provided in the legends of Fig. 6c–i, Extended Data Fig. 7b and Supplementary Figs. 9b and 13c. The experiments were independently repeated at least thrice.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The genome assembly T2T-pig1.0 and raw sequencing data generated in this study, including PacBio HiFi, ultralong ONT, MGI, Iso-seq and ChIP-seq data, can be accessed from the National Center for Biotechnology Information (<https://www.ncbi.nlm.nih.gov/>) under the BioProject accession PRJNA1139427. The whole-genome sequencing data for Ding'an and Wuzhishan are available in NCBI under the BioProject accessions PRJNA1161808 and PRJNA1161794, respectively. The haplotypic assemblies Hap1 and Hap2 for T2T-pig1.0 are available via figshare at <https://doi.org/10.6084/m9.figshare.28246343.v1>. Source data are provided with this paper.

Code availability

The custom scripts and codes used in this study are available via GitHub at <https://github.com/Hellohuangning/T2T-pig1.0> and Zenodo at <https://doi.org/10.5281/zenodo.17274480> (ref. 118).

References

60. Camacho, C. et al. BLAST+: architecture and applications. *BMC Bioinformatics* **10**, 421 (2009).
61. Chen, S., Zhou, Y., Chen, Y. & Gu, J. fastp: an ultra-fast all-in-one FASTQ preprocessor. *Bioinformatics* **34**, i884–i890 (2018).
62. Langmead, B. & Salzberg, S. L. Fast gapped-read alignment with Bowtie 2. *Nat. Methods* **9**, 357–359 (2012).
63. Burton, J. N. et al. Chromosome-scale scaffolding of de novo genome assemblies based on chromatin interactions. *Nat. Biotechnol.* **31**, 1119–1125 (2013).
64. Li, H. Minimap2: pairwise alignment for nucleotide sequences. *Bioinformatics* **34**, 3094–3100 (2018).
65. Marçais, G. & Kingsford, C. A fast, lock-free approach for efficient parallel counting of occurrences of k -mers. *Bioinformatics* **27**, 764–770 (2011).
66. Thorvaldsdóttir, H., Robinson, J. T. & Mesirov, J. P. Integrative Genomics Viewer (IGV): high-performance genomics data visualization and exploration. *Brief. Bioinform.* **14**, 178–192 (2013).
67. Hu, J. et al. NextPolish2: a repeat-aware polishing tool for genomes assembled using HiFi long reads. *Genomics Proteomics Bioinformatics* **22**, qzad009 (2024).
68. Hu, J., Fan, J., Sun, Z. & Liu, S. NextPolish: a fast and efficient genome polishing tool for long-read assembly. *Bioinformatics* **36**, 2253–2255 (2020).
69. Alonge, M. et al. Automated assembly scaffolding using RagTag elevates a new tomato system for high-throughput genome editing. *Genome Biol.* **23**, 258 (2022).
70. Li, H. & Durbin, R. Fast and accurate long-read alignment with Burrows-Wheeler transform. *Bioinformatics* **26**, 589–595 (2010).
71. Gel, B. & Serra, E. karyoPloteR: an R/Bioconductor package to plot customizable genomes displaying arbitrary data. *Bioinformatics* **33**, 3088–3090 (2017).
72. Zhou, Z. W. et al. GenomeSyn: a bioinformatics tool for visualizing genome synteny and structural variations. *J. Genet. Genomics* **49**, 1174–1176 (2022).
73. Simao, F. A., Waterhouse, R. M., Ioannidis, P., Kriventseva, E. V. & Zdobnov, E. M. BUSCO: assessing genome assembly and annotation completeness with single-copy orthologs. *Bioinformatics* **31**, 3210–3212 (2015).
74. Rhie, A., Walenz, B. P., Koren, S. & Phillippy, A. M. Merqury: reference-free quality, completeness, and phasing assessment for genome assemblies. *Genome Biol.* **21**, 245 (2020).
75. Quinlan, A. R. & Hall, I. M. BEDTools: a flexible suite of utilities for comparing genomic features. *Bioinformatics* **26**, 841–842 (2010).
76. Ou, S. et al. Benchmarking transposable element annotation methods for creation of a streamlined, comprehensive pipeline. *Genome Biol.* **20**, 275 (2019).
77. Chen, N. Using RepeatMasker to identify repetitive elements in genomic sequences. *Curr. Protoc. Bioinformatics* **25**, 4.10.1–4.10.14 (2004).
78. Iseric, H., Alkan, C., Hach, F. & Numanagic, I. Fast characterization of segmental duplication structure in multiple genome assemblies. *Algorithms Mol. Biol.* **17**, 4 (2022).
79. Pertea, M. et al. StringTie enables improved reconstruction of a transcriptome from RNA-seq reads. *Nat. Biotechnol.* **33**, 290–295 (2015).
80. Keilwagen, J. et al. Using intron position conservation for homology-based gene prediction. *Nucleic Acids Res.* **44**, e89 (2016).
81. Stanke, M., Diekhans, M., Baertsch, R. & Haussler, D. Using native and syntetically mapped cDNA alignments to improve de novo gene finding. *Bioinformatics* **24**, 637–644 (2008).
82. Cantarel, B. L. et al. MAKER: an easy-to-use annotation pipeline designed for emerging model organism genomes. *Genome Res.* **18**, 188–196 (2008).
83. Nawrocki, E. P. & Eddy, S. R. Infernal 1.1: 100-fold faster RNA homology searches. *Bioinformatics* **29**, 2933–2935 (2013).
84. Kalvari, I. et al. Rfam 14: expanded coverage of metagenomic, viral and microRNA families. *Nucleic Acids Res.* **49**, D192–D200 (2021).
85. Chan, P. P., Lin, B. Y., Mak, A. J. & Lowe, T. M. tRNAscan-SE 2.0: improved detection and functional classification of transfer RNA genes. *Nucleic Acids Res.* **49**, 9077–9096 (2021).
86. Lagesen, K. et al. RNAmmer: consistent and rapid annotation of ribosomal RNA genes. *Nucleic Acids Res.* **35**, 3100–3108 (2007).
87. Gasteiger, E., Jung, E. & Bairoch, A. SWISS-PROT: connecting biomolecular knowledge via a protein database. *Curr. Issues Mol. Biol.* **3**, 47–55 (2001).
88. Kanehisa, M. & Goto, S. KEGG: kyoto encyclopedia of genes and genomes. *Nucleic Acids Res.* **28**, 27–30 (2000).
89. Mistry, J. et al. Pfam: the protein families database in 2021. *Nucleic Acids Res.* **49**, D412–D419 (2021).
90. Jones, P. et al. InterProScan 5: genome-scale protein function classification. *Bioinformatics* **30**, 1236–1240 (2014).
91. Vollger, M. R., Kerpedjiev, P., Phillippy, A. M. & Eichler, E. E. StainedGlass: interactive visualization of massive tandem repeat structures with identity heatmaps. *Bioinformatics* **38**, 2049–2051 (2022).
92. Kokot, M., Dlugosz, M. & Deorowicz, S. KMC 3: counting and manipulating k -mer statistics. *Bioinformatics* **33**, 2759–2761 (2017).
93. Chen, J. et al. A complete telomere-to-telomere assembly of the maize genome. *Nat. Genet.* **55**, 1221–1231 (2023).
94. Hu, T. T. et al. The *Arabidopsis lyrata* genome sequence and the basis of rapid genome size change. *Nat. Genet.* **43**, 476–481 (2011).
95. Slotte, T. et al. The *Capsella rubella* genome and the genomic consequences of rapid mating system evolution. *Nat. Genet.* **45**, 831–835 (2013).
96. Ellinghaus, D., Kurtz, S. & Willhoeft, U. LTRharvest, an efficient and flexible software for de novo detection of LTR retrotransposons. *BMC Bioinformatics* **9**, 18 (2008).

97. Edgar, R. C. MUSCLE: multiple sequence alignment with high accuracy and high throughput. *Nucleic Acids Res.* **32**, 1792–1797 (2004).
98. Zhang, M., Yang, Q., Ai, H. & Huang, L. Revisiting the evolutionary history of pigs via de novo mutation rate estimation in a three-generation pedigree. *Genomics Proteomics Bioinformatics* **20**, 1040–1052 (2022).
99. Zheng, Z. et al. The origin of domestication genes in goats. *Sci. Adv.* **6**, eaaz5216 (2020).
100. Li, X. et al. Whole-genome resequencing of wild and domestic sheep identifies genes associated with morphological and agronomic traits. *Nat. Commun.* **11**, 2815 (2020).
101. Chen, C. et al. TBtools: an integrative toolkit developed for interactive analyses of big biological data. *Mol. Plant* **13**, 1194–1202 (2020).
102. Johnson, T. et al. Short communication: identification of the pseudoautosomal region in the Hereford bovine reference genome assembly ARS-UCD1.2. *J. Dairy Sci.* **102**, 3254–3258 (2019).
103. Wu, T. et al. clusterProfiler 4.0: a universal enrichment tool for interpreting omics data. *Innovation (Camb.)* **2**, 100141 (2021).
104. Stamatakis, A. RAxML-VI-HP: maximum likelihood-based phylogenetic analyses with thousands of taxa and mixed models. *Bioinformatics* **22**, 2688–2690 (2006).
105. Yang, Z. PAML 4: phylogenetic analysis by maximum likelihood. *Mol. Biol. Evol.* **24**, 1586–1591 (2007).
106. Lu, Y. Y. et al. CAFE: aCcelerated Alignment-FrEe sequence analysis. *Nucleic Acids Res.* **45**, W554–W559 (2017).
107. Tang, H. et al. JCVI: a versatile toolkit for comparative genomics analysis. *iMeta* **3**, e211 (2024).
108. Bolger, A. M., Lohse, M. & Usadel, B. Trimmomatic: a flexible trimmer for Illumina sequence data. *Bioinformatics* **30**, 2114–2120 (2014).
109. Jung, Y. & Han, D. BWA-MEME: BWA-MEM emulated with a machine learning approach. *Bioinformatics* **38**, 2404–2413 (2022).
110. Danecek, P. et al. Twelve years of SAMtools and BCFtools. *GigaScience* **10**, giab008 (2021).
111. McKenna, A. et al. The genome analysis toolkit: a MapReduce framework for analyzing next-generation DNA sequencing data. *Genome Res.* **20**, 1297–1303 (2010).
112. Letunic, I. & Bork, P. Interactive Tree Of Life (iTOL) v5: an online tool for phylogenetic tree display and annotation. *Nucleic Acids Res.* **49**, W293–W296 (2021).
113. Patterson, N., Price, A. L. & Reich, D. Population structure and eigenanalysis. *PLoS Genet.* **2**, e190 (2006).
114. Wang, J. Fast and accurate population admixture inference from genotype data from a few microsatellites to millions of SNPs. *Heredity* **129**, 79–92 (2022).
115. Wang, K., Li, M. & Hakonarson, H. ANNOVAR: functional annotation of genetic variants from high-throughput sequencing data. *Nucleic Acids Res.* **38**, e164 (2010).
116. Li, T. F. et al. Smad3-deficient chondrocytes have enhanced BMP signaling and accelerated differentiation. *J. Bone Miner. Res.* **21**, 4–16 (2006).
117. Yoon, D. S. et al. RUNX2 stabilization by long non-coding RNAs contributes to hypertrophic changes in human chondrocytes. *Int. J. Biol. Sci.* **19**, 13–33 (2023).
118. Luo, Y., Huang, N., Zha, C., Jia, S. & Fang, M. Scripts for T2T-pig1.0. Zenodo <https://doi.org/10.5281/zenodo.17274480> (2025).

Acknowledgements

We thank N. Li, Z. H. Jiang, L. Andersson, C. X. Wu, K. Li, G. L. Liu and S. Y. Tan for their valuable discussions and suggestions. This work was supported by the National Natural Science Foundation of China (U22A20508 to M.-Y.F.), the National Key Research and Development Program of China (2023YFF1001000 to M.-Y.F. and 2022YFF1003402 to S.-G.J.), the Pinduoduo-China Agricultural University Research Fund (PC2024A01003 to M.-Y.F.) and the China Agriculture Research System of MOF and MARA (CARS-36 to M.-Y.F.).

Author contributions

M.-Y.F. conceived of the project. M.-Y.F. and S.-G.J. supervised the study. Y.-B.L. and N.H. performed genome assembly and data analysis. C.-W.Z. participated in data analysis and drafting of the paper. Z.C., R.-P.S. and F.W. provided partial help and financial support for the study. Y.-B.W. was involved in data analysis. X.-Y.Y. and P.-X.X. were involved in the interpretation and plotting of the results. Q.X., Y.-B.L. and L.-M.Z. performed the blood and tissue collection for the Wuzhishan pig. L.-X.Y. completed the isolation of porcine chondrocytes and functional validation of candidate genes. S.-G.J., Y.-B.L. and M.-Y.F. wrote the paper.

Competing interests

All authors declare no competing interests.

Additional information

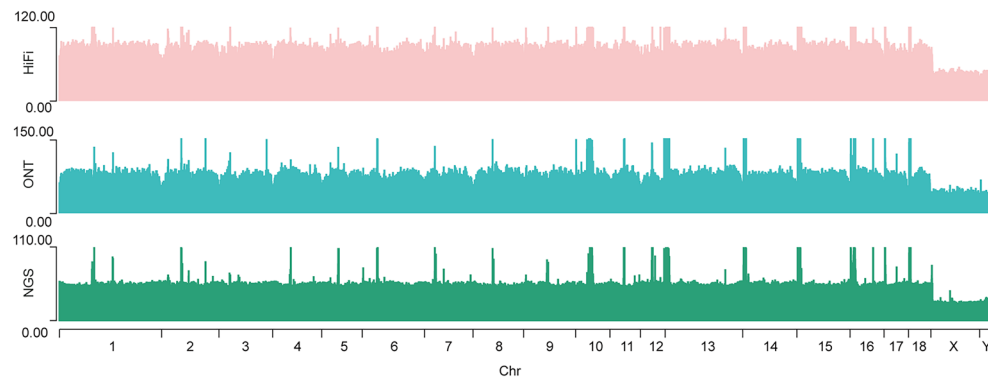
Extended data is available for this paper at <https://doi.org/10.1038/s41588-025-02433-6>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41588-025-02433-6>.

Correspondence and requests for materials should be addressed to Shan-Gang Jia or Mei-Ying Fang.

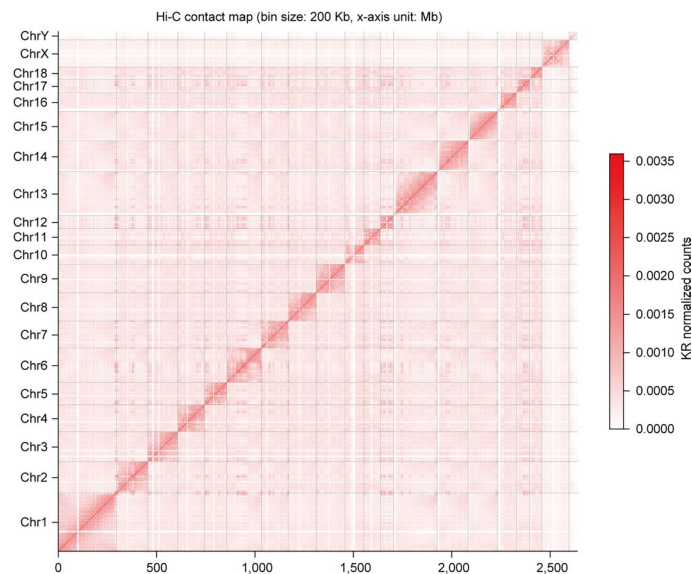
Peer review information *Nature Genetics* thanks Lingzhao Fang and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available.

Reprints and permissions information is available at www.nature.com/reprints.



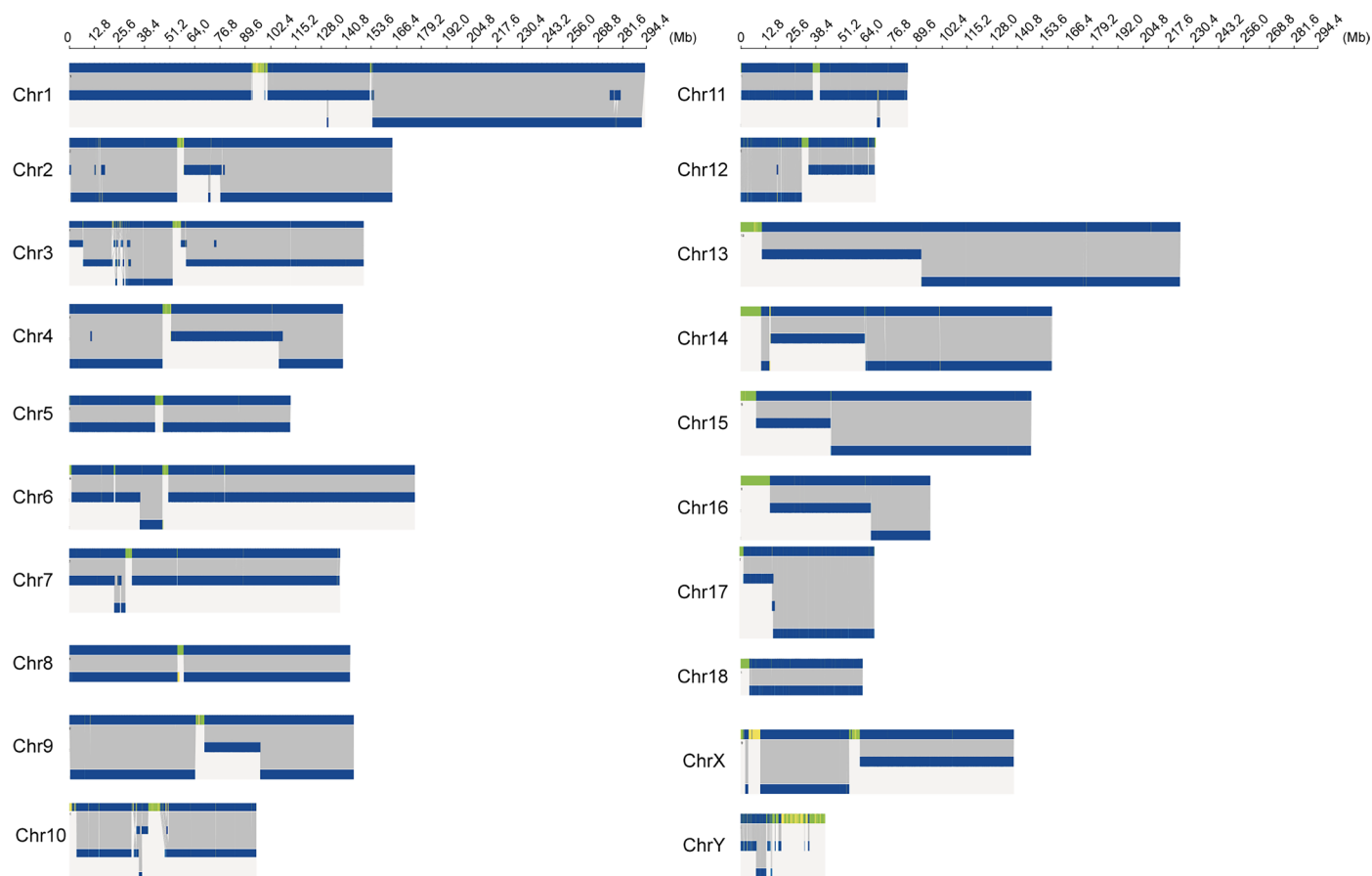
Extended Data Fig. 1 | Coverage of mapped reads across all 20 chromosomes of T2T-pig1.0. Coverage was calculated using mapped reads from three sequencing technologies: next-generation sequencing (NGS) for short MGI reads, PacBio HiFi for long reads, and Oxford Nanopore Technology (ONT) for long reads. The y-axis indicates read coverage (in counts), while the x-axis represents the

20 chromosomes in 1-Mb bins. Regions with unusually high coverage often correspond to the centromeric regions with enriched repetitive sequences. Additionally, the coverages of the X and Y chromosomes are approximately half of those on the autosomes.

**Extended Data Fig. 2 | Hi-C interaction for the 20 chromosomes of T2T-pig1.0.**

The Hi-C contact map displays chromatin interaction frequencies across the chromosomes in T2T-pig1.0 genome assembly. The map was generated using a bin size of 200 kb, with chr1–chr18, chrX and chrY listed in the ordered columns on the x-axis and the ordered rows on the y-axis. A scale is added on the x-axis in megabases (Mb). Each pixel's intensity reflects KR (Knight–Ruiz) normalized

interaction counts, as indicated by the color scale bar. Strong diagonal signals represent intrachromosomal interactions, indicating high contiguity and structural completeness of the assembly. Cross-chromosomal interaction signals (off-diagonal) are comparatively weaker, consistent with expected patterns of chromatin organization.



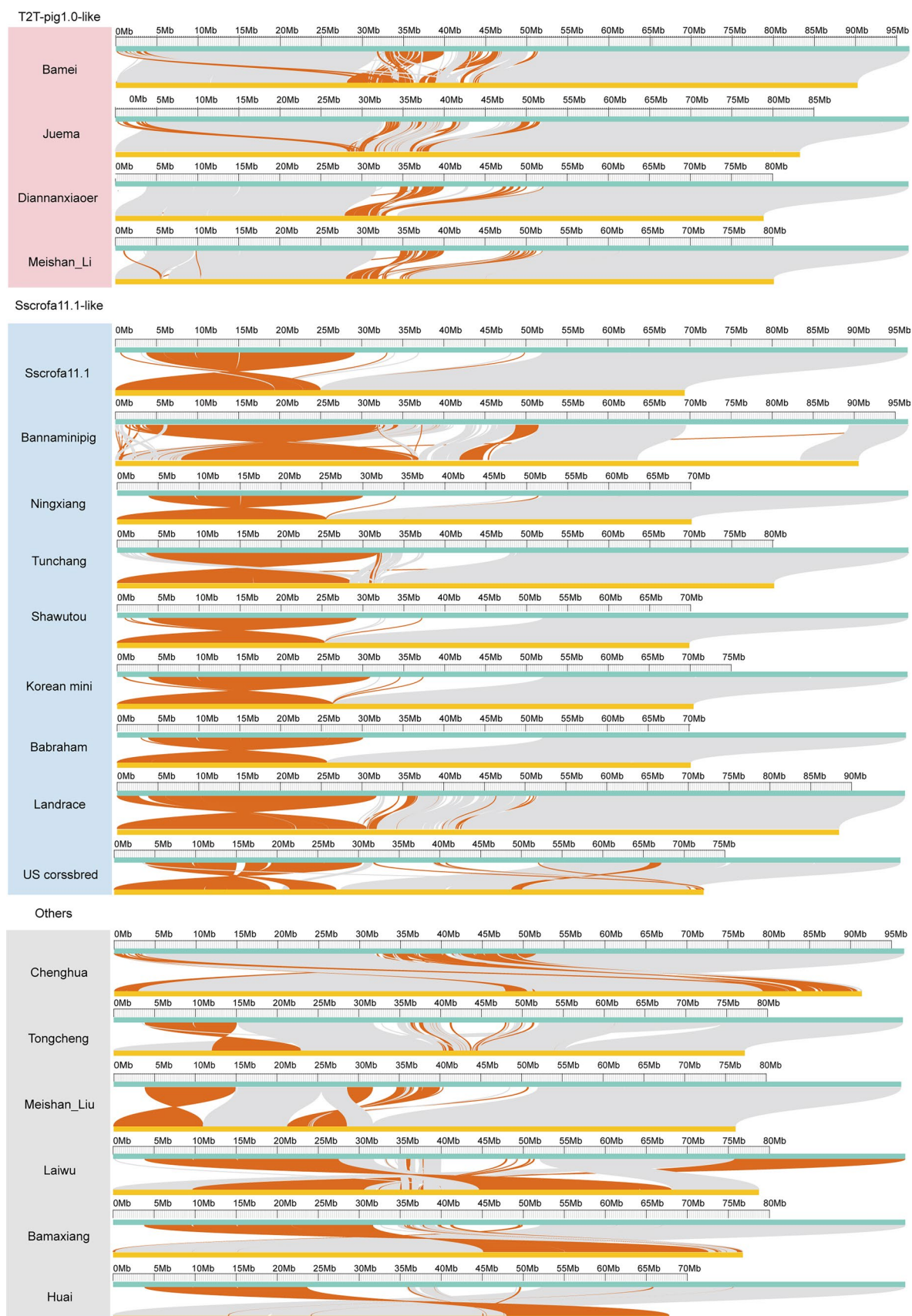
Extended Data Fig. 3 | Alignment of Bionano DLS optical maps against in silico maps of the T2T-pig1.0 genome assembly. The top bars for T2T-pig1.0 chromosomes and the bottom bars for Bionano optical maps are linked by the gray blocks or lines. The blue color indicates the matched regions from T2T-pig1.0 chromosomes and Bionano optical maps. The yellow region in T2T-pig1.0 represents the break of collinearity between T2T-pig1.0 and

Bionano optical maps, while the green region indicates no enzyme labeling sites in Bionano optical maps. The centromeric region is rich in repetitive sequences and lack of the enzyme marker sites for Bionano. The right end on the q arm of Y chromosome is highly enriched in repetitive sequences (SatY3), which results into a lot of yellow and green colors.



Extended Data Fig. 4 | Collinearity between the T2T-pig1.0 and Sscrofa11.1 genomes. The collinear regions between T2T-pig1.0 (bottom) and Sscrofa11.1 (top) are linked by gray blocks or lines. The PURs are shown on the chromosomes of T2T-pig1.0 in yellow. The dark blue and light blue triangles indicate the

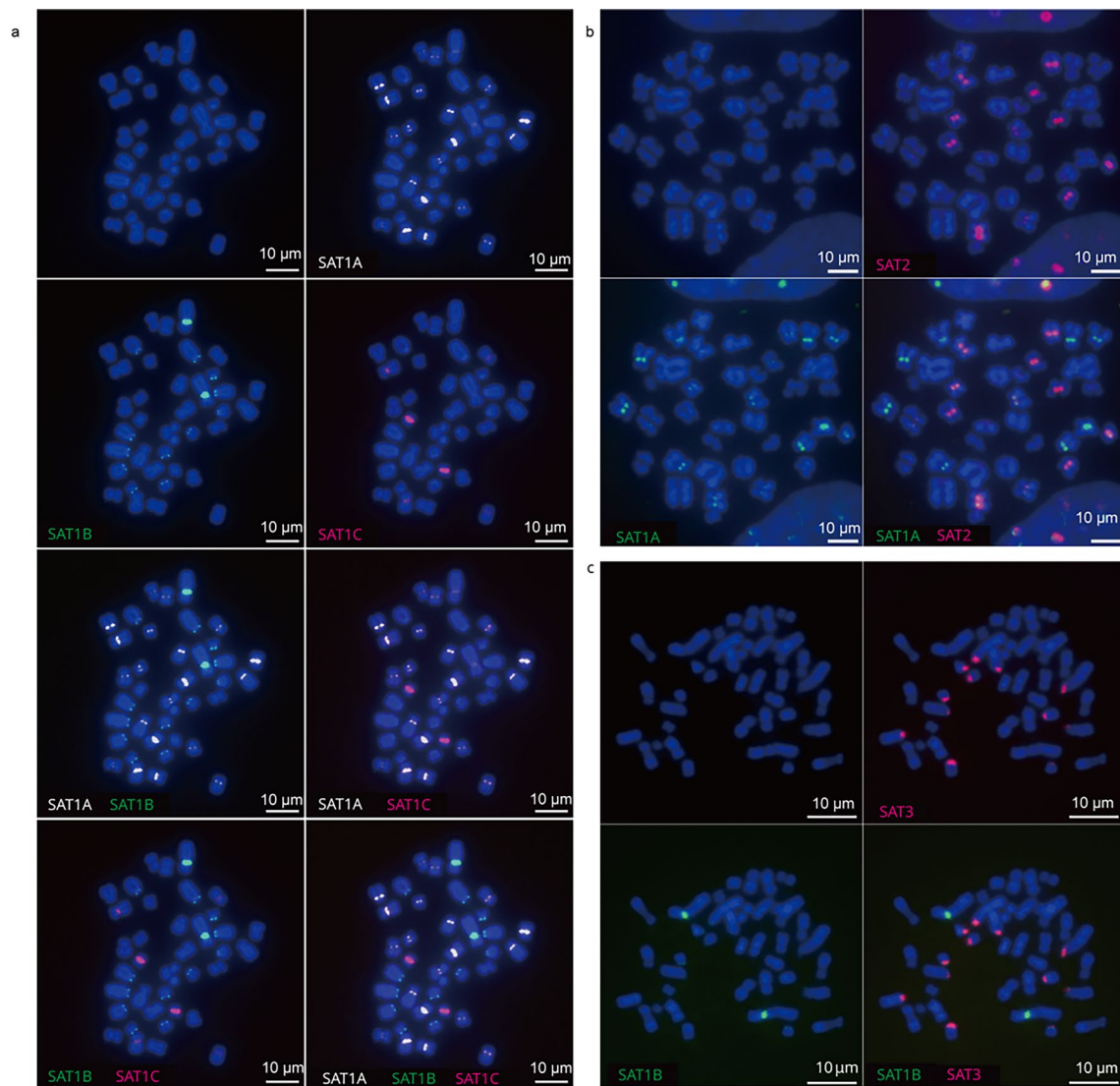
presence of telomere regions in T2T-pig1.0 and Sscrofa11.1, respectively. The red and gray triangles indicate the presence of centromeric regions in T2T-pig1.0 and Sscrofa11.1, respectively. The telomeric sequences and centromeres information for Sscrofa11.1 was derived from the previous paper (ref. 7).



Extended Data Fig. 5 | See next page for caption.

Extended Data Fig. 5 | A large SV on chromosome 10 in the available pig genome assemblies. The top bars in cyan represent T2T-pig1.0, with yellow bars for other pig genome assemblies. The orange blocks or lines represent inversions, while the gray blocks or lines link the syntenic regions. A large SV (chr10–SV1, chr10: 0.94–29.16 Mb) between T2T-pig1.0 and Sscrofa11.1 is confirmed in the comparisons with other pig genome assemblies. There are three groups for this large SV, that is, T2T-pig1.0-like, Sscrofa11.1-like and others, and it indicates a true SV for chr10–SV1 between T2T-pig1.0 and Sscrofa11.1 rather than an assembly error. The NCBI accession numbers for the pig genome assemblies are shown as follows: Bamei, [GCA_030704935.1](#); Juema, [GCA_040869115.1](#);

Diannanxiaoer, GWHACDT000000000 (CNCB database, <https://ngdc.cncb.ac.cn/gwh/>); Meishan_Li, [GCA_017957985.1](#); Sscrofa11.1, [GCA_000003025.6](#); Bannaminipig, [GCA_041937265.1](#); Ningxiang, [GCA_020567905](#); Korean mini, [GCA_039654815.1](#); Babraham, [GCA_031225015.1](#); Landrace, [GCA_001700215.1](#); US crossbred, [GCA_002844635.1](#); Chenghua, [GCA_037447515.1](#); Tongcheng, GWHDTXM000000000 (CNCB); Meishan_Liu, GWHDTXE000000000 (CNCB); Laiwu, GWHDTXD000000000 (CNCB); Bamaxiang, [GCA_007644095.1](#); Huai, [GCA_041729645.1](#) and two figshare datasets for Tunchang and Shawutou (<https://figshare.com/s/155e92372e5e8e7b4e0a>).



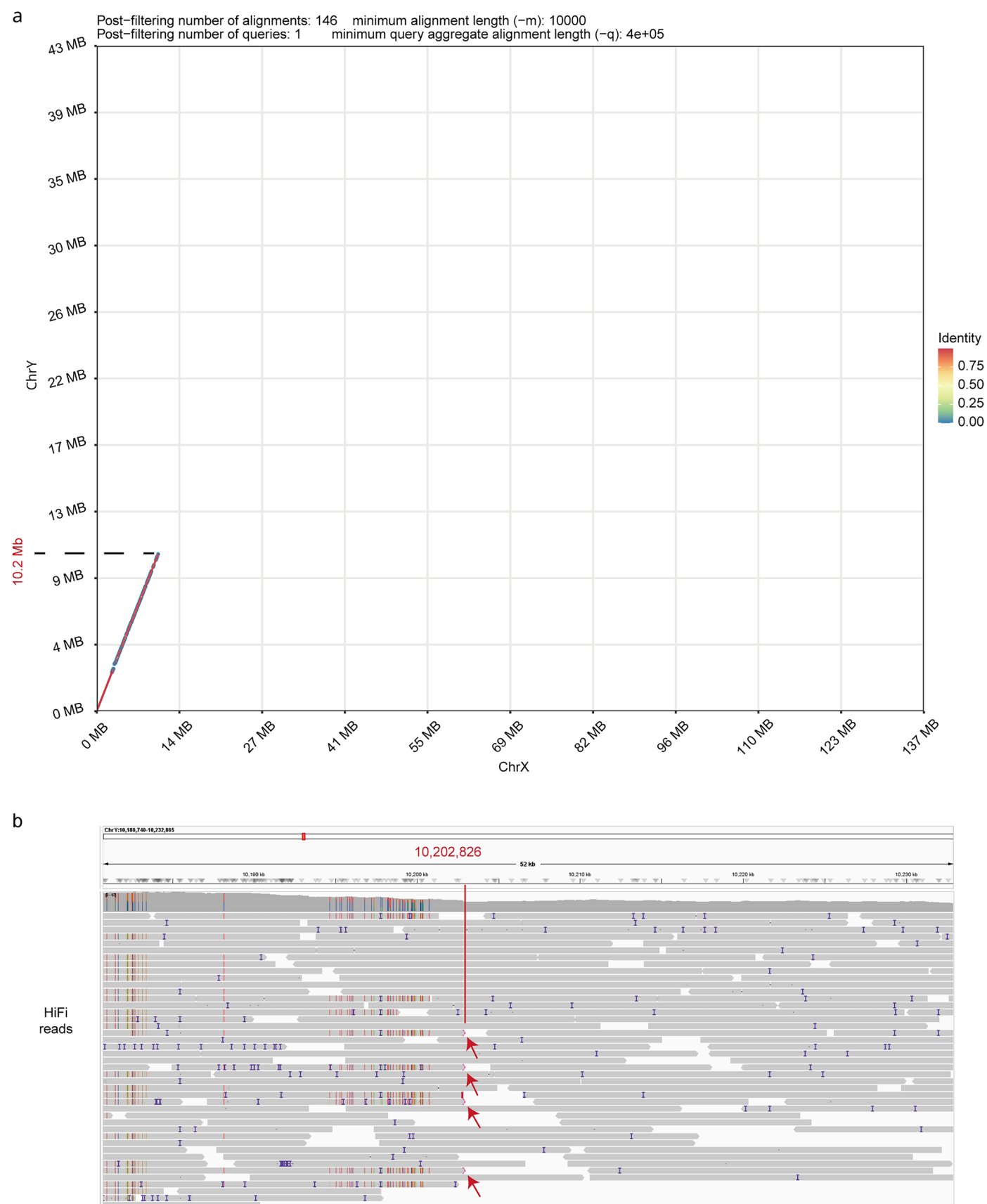
Extended Data Fig. 6 | Fluorescence in situ hybridization (FISH) results for SAT1, SAT2, and SAT3 probes. a, FISH images for the probes of SAT1A (white), SAT1B (green), and SAT1C (pink), which are included in SAT1. **b,** Probes of SAT2 (pink) and SAT1A (green) were hybridized on the pig chromosomes. **c,** Probes of SAT3 (pink) and SAT1B (green) and their hybridization signals are shown on

telocentric chromosomes. Scale bar = 10 μm. The primary fibroblast cells were from the Wuzhishan pigs. The SAT1, SAT2, and SAT3 are synthesized and their complete sequences are included in Supplementary Table 7. All the experiments were repeated independently at least thrice with similar results.



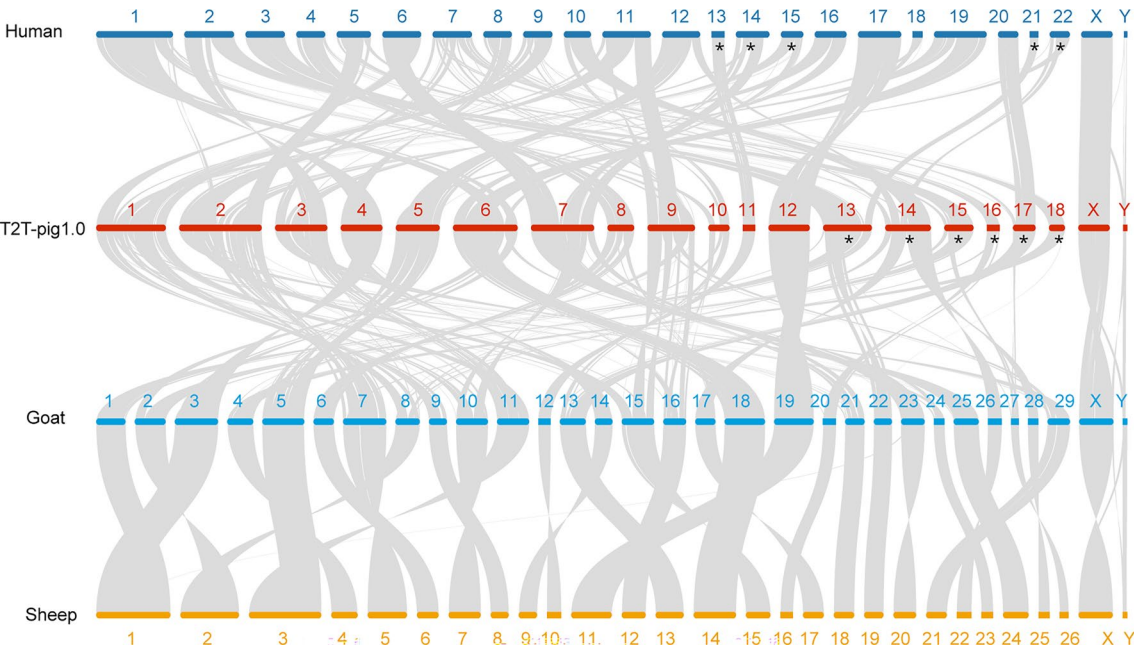
Extended Data Fig. 7 | LTR insert time and distribution differences between centromeric and non-centromeric regions. **a**, The LTRs highlighted by insert time in colors across the chromosomes in the T2T genome assemblies of pigs (T2T-pig1.0), humans (T2T-CHM13v2.0), sheep (T2T-sheep1.0) and goats (T2T-goat1.0). **b**, The differences of LTR insert time between the centromeric and non-centromeric regions in T2T-pig1.0. The young LTRs are accumulated in the centromeric regions (left), and the chromosomal distribution of LTR insertion time is compared between the centromeric (LTR_centromere) and non-centromeric (LTR_non-centromere) regions in T2T-pig1.0. Rare LTR insertion is found in the telocentric chromosomes (chr13–chr18). Box plots show the 25th (top edges) to 75th (bottom edges) percentiles with median lines, and whiskers

extending to 1.5× the interquartile range. A hash (#) indicates no LTR insert in the centromeric region of chr17. The LTR numbers in centromeric and non-centromeric regions are shown for the 20 chromosomes as follows: chr1 (8 versus 1,198), chr2 (17 versus 854), chr3 (15 versus 773), chr4 (12 versus 646), chr5 (34 versus 607), chr6 (3 versus 1,011), chr7 (19 versus 716), chr8 (1 versus 552), chr9 (35 versus 640), chr10 (71 versus 585), chr11 (41 versus 376), chr12 (2 versus 483), chr13 (1 versus 820), chr14 (1 versus 701), chr15 (1 versus 551), chr16 (3 versus 292), chr17 (0 versus 336), chr18 (1 versus 257), chrX (2 versus 842), chrY (384 versus 482). *P* values were calculated by the two-sided Wilcoxon rank-sum test with continuity correction using the `wilcox.test` function in R. Comparative analyses were performed separately for each chromosome. **P* < 0.05; ***P* < 0.01.



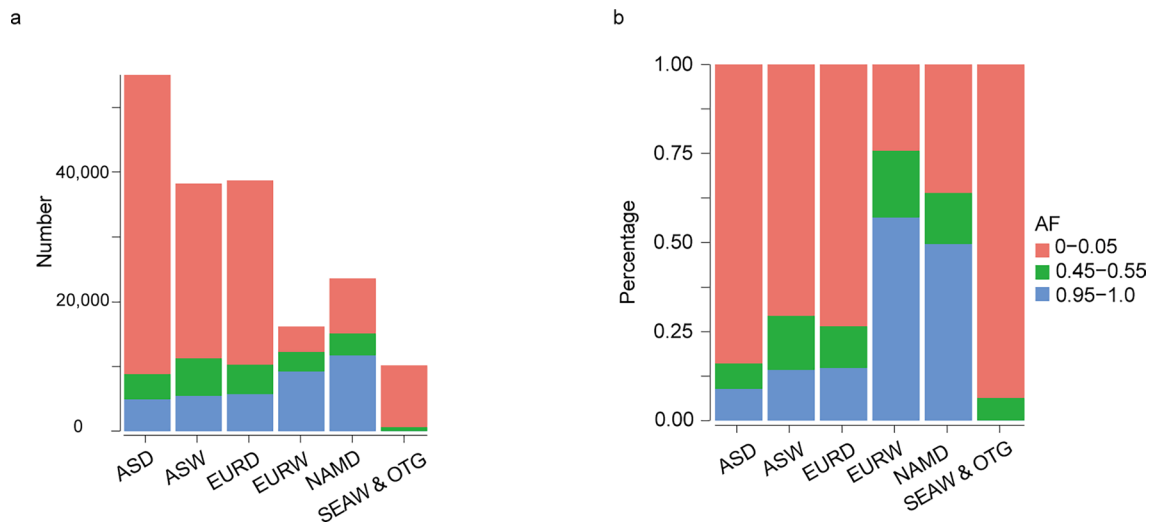
Extended Data Fig. 8 | PAR identification of X and Y chromosomes in T2T-pig1.0. **a**, Collinearity diagrams of the X and Y chromosomes. The dotted lines are colored based on the sequence identity with color key in the right, and indicate the putative PAR boundary at ~10.2 Mb of Y chromosome in T2T-pig1.0.

b, The HiFi reads coverage in PAR on chromosome Y. The red solid line indicates the hypothetical boundary of the PAR. The red arrows indicate the clipped HiFi reads probably from the maternal X-chromosome's PAR, whose right ends cannot be mapped to the PAR on chromosome Y.



Extended Data Fig. 9 | Gene syntenies analysis of the T2T genomes of humans, sheep, goats, and pigs. The gene syntenies are shown among the T2T genomes of pigs, humans, sheep, and goats, whose accession numbers in NCBI are shown as follows: GCF_009914755.1 (human), [GCA_040805955.1](#) (sheep), and

[GCA_041735815.1](#) (goat). The asterisks indicate the telocentric chromosomes in humans and T2T-pig1.0. In goats, all chromosomes are telocentric, whereas in sheep, only chromosomes 1, 2, and 3 are metacentric. The homologous genes among the four species are linked by gray lines.



Extended Data Fig. 10 | SNPs in PURs and their allele frequency across different geographical populations. **a**, The SNPs in PURs are shown in the different geographical populations. ASD, Asian domestic pig; ASW, Asian wild pig; EURD, European domestic pig; EURW, European wild pig; NAMD, North American

domestic pig; OTG (outgroup), *Phacochoerus africanus*; and SEAW, Southeast Asian warty pig. **b**, The percentages of allele frequencies (AF) are compared in these populations.

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a Confirmed

- ☐ ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
- ☐ ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☐ ☒ The statistical test(s) used AND whether they are one- or two-sided
Only common tests should be described solely by name; describe more complex techniques in the Methods section.
- ☒ ☐ A description of all covariates tested
- ☒ ☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☐ ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
- ☐ ☒ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
- ☒ ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☒ ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☒ ☐ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection PacBio Sequel II platform produced HiFi SIRT long reads. Nanopore PromethION platform produced ONT ultra-long reads. PacBio Sequel II platform produced ISO-seq reads. MGISEQ-T7 platform produced NGS reads. Illumina NovaSeq 6000 platform produced ChIP-seq reads.

Data analysis All softwares used in the present study are publicly available from the Internet, and the corresponding versions were described in detail on the section of Methods. Details as follows: Hifiasm (v0.19.4), Nextdenovo (v2.5.2), fastp (v0.23.4), Bowtie2 (v2.4.2), LACHESIS, BLASTN (v2.10.0), Minimap2 (v2.23), Jellyfish (v2.3.0), IGV (v2.13), RagTag (v2.1.0), Nextpolish2 (v0.1.0), BWA mem (v0.7.17-r1188), KaryoPloteR package (v1.8.4), BUSCO (v5.1.3), Merqury (v1.3), EDTA (v2.2), RepeatMasker (v4.1.4), BISER (v1.4), STAR (v2.7.9a), StringTie (v2.2.1), GeMoMa (v1.9), AUGUSTUS (v3.4.0), MAKER(v3.01.03), Infernal (v1.1.4), Rfam (v14.9), tRNAscan-SE (v2.0.12), RNAmme (v1.2), BLASTP (v2.10.0), InterproScan (v5.58), BEDTools (v2.30.0), StainedGlas (v0.5), KMC (v3.1.1), pbmm2 (v1.13.0), pb-CpG-tools (v2.3.2), cellSens Dimension (v1.9), Muscle (v3.8.31), TBtools (v2.056), OrthoFinder (v2.5.3), clusterProfiler (v3.19), RAXML (v1.2.2), pbsv (v2.9.0), CAFE (v1.1), JCVI (1.4.25), cuteSV (v2.0.2), Sniffles (v2.0.7), SAMtools (v1.16), SURVIVOR (v1.0.7), FastQC (v0.11.9), Trimmomatic (v0.39), SAMtools (v1.16), GATK (v4.3), VCFtools (v0.1.16), Delly (v0.8.7), Manta (v1.6.0), Smoove (v0.2.8), iTOL (v6.8.1), PHYLIP (v3.697), ADMIXTURE (v1.3.0), SlideView(v2.8.0), ANNOVAR, R (v4.4.2), GraphPad Prism (v9.5).

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

The genome assembly T2T-pig1.0 and raw sequencing data generated in this study, including PacBio HiFi, ultralong ONT, MGI, Iso-seq and ChIP-seq data, can be achieved from the National Center for Biotechnology Information (<https://www.ncbi.nlm.nih.gov/>) under the BioProject accession number PRJNA1139427. The whole genome sequencing data for Ding'an and Wuzhishan are available in NCBI under the BioProject accession numbers PRJNA1161808 and PRJNA1161794 respectively. The haplotypic assemblies Hap1 and Hap2 for T2T-pig1.0 are available in the Figshare website (<https://doi.org/10.6084/m9.figshare.28246343.v1>).

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

Reporting on race, ethnicity, or other socially relevant groupings

Population characteristics

Recruitment

Ethics oversight

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	To explore important genes associated with pig body stature, we selected Wuzhishan pigs as a sample for porcine genome assembly and used the genetic diversity of 407 individuals from worldwide, of which 50 individuals from Wuzhishan and Ding'an pigs were provided by this study. The 407 individuals used in this study were selected on the basis of pedigree, origin, differences in breed characteristics and breeding availability. This population is a representative sample of pig genetic resources, and we hope to discover some genes related to body stature breeding through this population.
Data exclusions	Only quality filtered HiFi reads and ONT ultra-long reads were used for genome assembly.
Replication	Replications for each experiment were clearly stated in figure legends or Methods section. Replications of repeated experiments were evaluated by proper statistic analyses and confirmed to be successful.
Randomization	The sampling process for DNA sequencing, RNA-seq, Iso-seq and ChIP-seq was randomly conducted.
Blinding	Blinding is not necessary for T2T genome sequencing and assembly. No blinding should not affect interpretation as all our experiment measures were objective.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☐	☒ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	CENP-A (Enzo Life Sciences, ADI-KAM-CC006-E).
Validation	The antibody against CENH3 was validated to be available according to a previous report (A mitotic chromatin phase transition prevents perforation by microtubules: M.W.G. Schneider, et al.; Nature 609, 183 (2022)), in which the anti-CENH3 antibody used is same as we used here.

Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s)	Primary chondrocytes were isolated from the thoracic intervertebral cartilage and epiphyseal cartilage of newborn piglets.
Authentication	Primary chondrocytes were characterized by immunofluorescence analysis with a COL2A1 antibody (Boster Bio, PA2141-1).
Mycoplasma contamination	The cell lines were not tested for mycoplasma contamination.
Commonly misidentified lines (See ICLAC register)	No commonly misidentified cell lines used in the study.

Animals and other research organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research, and [Sex and Gender in Research](#)

Laboratory animals	In this study, fresh whole blood and tissue samples were collected from a 120-day-old healthy boar from the Wuzhishan Pig Breeding Farm of the Hainan Academy of Agricultural Sciences (Haikou, Hainan Province).
Wild animals	No wild animals were involved in this study.
Reporting on sex	Wuzhishan Pig: male.
Field-collected samples	There are no field-collected samples in this study.
Ethics oversight	All animal procedures were performed in accordance with the guidelines for animal experiments approved by the China Agricultural University Institutional Animal Care and Use Committee (AW31904202-1-1).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Plants

Seed stocks	Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.
Novel plant genotypes	Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.
Authentication	Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.

ChIP-seq

Data deposition

- ☒ Confirm that both raw and final processed data have been deposited in a public database such as [GEO](#).
- ☒ Confirm that you have deposited or provided access to graph files (e.g. BED files) for the called peaks.

Data access links

May remain private before publication.

<https://dataview.ncbi.nlm.nih.gov/object/PRJNA1139427?reviewer=1hgjmro14tu55g4qshfe38s7pq>

Files in database submission

SRR30630686, SRR30630651

Genome browser session
(e.g. [UCSC](#))

No longer applicable.

Methodology

Replicates

No experimental replicates.

Sequencing depth

Raw total reads: 68,040,844, mapped and paired reads: 58,992,868, unmapped reads: 9,047,976.

Antibodies

CENP-A (Enzo Life Sciences, ADI-KAM-CC006-E).

Peak calling parameters

After trimming with fastp (v0.23.1), the clean reads were aligned to T2T-pig1.0 via Bowtie2 (v2.4.2) with the parameters “--very-sensitive --no-mixed --no-discordant -k 10”. The read depth for ChIP enrichment in 5-kb windows was determined using BEDTools (v2.30.0) and visualized with the KaryoPloteR package (v1.8.4). Heatmaps of sequence identity for centromeric repeats were created using StainedGlas (v0.5), with color coding representing percentage identity.

Data quality

All twenty centromeres in pig were successfully identified.

Software

Fastp (v0.23.1), Bowtie2 (v2.4.2), BEDTools (v2.30.0), KaryoPloteR package (v1.8.4), StainedGlas (v0.5).